

A reusable solvent-response framework for molecular hydration prediction

Gal Oren, Boris Fain and Michael Levitt

Stanford University, Stanford, California, USA

Abstract

Hydration free energy connects molecular structure to partitioning, binding and reactivity, but resolving solvent interactions afresh for each molecule is costly. We ask whether calculated and measured solvent responses can instead be learned as reusable molecular features. SolvAI predicts 15 descriptors from six external sources and passes them, with structural features, to an experimentally supervised hydration predictor. The response representation is independent of the endpoint architecture; inference starts from a SMILES string and requires no new solvent calculation. We evaluated tree ensembles, neural networks, graph models, pretrained molecular encoders, TabM and TabPFN under matched data partitions. Adding responses lowers ARROW-85 and External-220 MAE for ExtraTrees (0.303 to 0.202 and 1.532 to 1.153 kcal mol⁻¹) and a MolFormer neural endpoint (0.465 to 0.273 and 1.332 to 1.111 kcal mol⁻¹). Message passing and TabM also benefit on the external cohort, whereas TabPFN has no resolved response increment there. Several response gains extend to the nested 97-molecule subset absent from all six supervised response-source tables; membership in pretrained encoder corpora was not audited. No endpoint has the lowest error in every evaluation. TabPFN reaches 0.199 kcal mol⁻¹ on ARROW and 1.318 on a separate exploratory size holdout, but the latter comparison uses model-specific input recipes. Predictions on unlabelled peptide series change with the endpoint but do not establish physical accuracy. Solvent-response descriptors can thus be reused across model families, while interpolation, external transfer and size generalization remain distinct tests.

Introduction

Hydration free energy is the reversible work of transferring a molecule from the gas phase into water. It connects molecular structure to partitioning, binding and reactivity, and is a stringent test of how a model balances electrostatics, polarization, dispersion, repulsion and solvent reorganization. Experimental values are available for many small neutral compounds, notably through FreeSolv, but predicting them with comparable precision remains difficult.^{1;2}

Physical simulation approaches the problem by resolving the solvent response. In an alchemical calculation the solute–water interaction is varied along a coupling path, and the resulting ensemble averages are integrated or combined statistically.³ More detailed Hamiltonians improve the intermolecular interaction; path-integral molecular dynamics (PIMD) can additionally represent nuclear delocalization, a nuclear quantum effect (NQE).⁴ These ingredients can be accurate, but they demand repeated sampling for each new molecule. The ARROW simulation study made this accuracy–cost relation unusually sharp: on 85 neutral solutes, classical ARROW has an exactly reconstructed mean absolute error (MAE) of 0.785 kcal mol⁻¹, whereas eight-bead PIMD (PIMD8) reduces it to 0.205 kcal mol⁻¹.⁵ Subsequent learned-potential work reproduced ARROW water-pair interactions and quantum corrections within simulation, showing that parts of this description are learnable while retaining trajectory-based inference.^{6;7}

Data-driven solvation models offer a way to reuse information across molecules. Direct predictors map a graph, fingerprint or three-dimensional structure to a free energy, while Δ -learning and multi-fidelity models transfer information between calculated and experimental targets.^{8–10} Graph models can also replace a particular implicit-solvent calculation by learning its output directly from molecular structure.¹¹ Physically inspired molecular descriptors can improve hydration prediction when they encode electrostatics, polarizability, accessible surface and conformational response.^{12;13} Other models retain physical computation at prediction time: hybrid alchemical/machine-learning (ML) methods correct a simulated free energy, ML-PCM learns from continuum-model components, 3D-RISM outputs have served as hydration thermodynamic descriptors, and implicitly perturbed Hamiltonians represent a molecule through its response to real or hypothetical solvents.^{14–17} These studies show the value of solvent-conditioned representations; in these latter approaches, an alchemical, continuum, integral-equation or ensemble calculation is still performed for a new molecule.

A complementary line of work learns the physical descriptor itself. The ml-QM-GNN architecture predicts

quantum-chemical atom and bond descriptors before using them in a reactivity model.¹⁸ Broader evaluations find that such descriptors help most when they are relevant to the endpoint and can be recovered accurately, and that a surrogate’s hidden representation can sometimes be more useful than its explicit predicted values.^{19;20} Most directly for hydration, physically inspired solute descriptors have been predicted from molecular graphs and passed to a second free-energy model, eliminating their per-query electronic structure calculation.²¹ These developments establish the major architectural ingredients. Combining predictions from multiple chemical models is also well established, as illustrated by the CATMoS consensus models for acute toxicity.²² Our question concerns the information being transferred: do complementary descriptions of solvent response remain useful for hydration prediction when each is inferred from structure, and how far does their contribution persist outside the chemistry used to fit the final predictor?

SolvAI is a solvent-response representation coupled to a learned hydration endpoint, not a particular endpoint algorithm (Fig. 1). Six sources provide 15 targets: calculated transfer free energies, empirical solute properties and summaries of how water changes conformer energies. Separate molecular surrogates predict those targets from structure and are then frozen. We use *response descriptor* for one of these predicted quantities; the term does not imply a derivative with respect to an applied field.

We evaluate the same response representation with tree ensembles, residual and branched neural networks, message-passing graphs, pretrained molecular encoders, TabM and TabPFN. Each fixed configuration has a matched fit without responses, holding its data and selected input recipe fixed. These paired comparisons are distinct from asking which endpoint has the lowest absolute error. We examine both questions on ARROW-85 and an external 220-molecule cohort, then compare eight configurations on a separate size holdout. Additional experiments examine chemical separation, source composition, molecule-shuffled responses and repeated partitions in the ExtraTrees configuration; they were not run for all endpoints. The architecture search is retrospective and is reported separately from the prespecified controls. This design asks whether a useful response signal persists across learners, and where endpoint choice changes interpolation, external transfer or behavior outside the training-size range.

SolvAI: learning solvent response before hydration free energy

Two learning tasks with different supervision

For a neutral molecule represented by a SMILES string x , the target is its experimental gas-to-water transfer free energy $y = \Delta G_{\text{hyd}}$ near 298 K. We first train separate molecular models on external source datasets. Each model predicts one or more source quantities, such as a continuum-solvent free energy or a hydrogen-bonding parameter. These models are *surrogates*: after training, they approximate the source quantities without repeating the source calculation or measurement. Their weights are frozen before learning the experimental hydration predictor.

Writing the frozen surrogate predictions as $\hat{r}_j(x)$, our model is

$$\hat{\mathbf{r}}(x) = (\hat{r}_1(x), \dots, \hat{r}_{15}(x)), \quad \hat{y}(x) = h([\mathbf{s}(x), \hat{\mathbf{r}}(x)]), \quad (1)$$

where $\mathbf{s}(x)$ denotes structural features; the standard tabular input contains a 2,048-bit Morgan fingerprint and 217 RDKit two-dimensional descriptors. The endpoint h learns how the predicted source quantities relate to experiment, rather than assuming that their sum is a physical free energy. We vary h across the endpoint families in Table 2; graph and pretrained-encoder variants can also change the structural representation. The same 15 frozen source predictions are available to every response-augmented fit. All are predicted at deployment, including those whose source targets were experimental. One of the evaluated endpoints averages three ExtraTrees regressors.²³

What the response descriptors represent

The six sources describe different aspects of solvation (Table 1; Fig. 2a). Continuum-solvent calculations summarize the energetic response of a solvent model; Abraham parameters describe empirical interaction tendencies; corrected explicit- and implicit-solvent estimates combine a physical prediction with a learned experimental residual. The conformational source summarizes both the redistribution of conformer energies in water and the variation of solvation energy across conformers. These are complementary targets, not successive levels on a single fidelity scale. The source methods are described in the COSMOtherm, SoluteML, OpenFF Sage, GBn2, SMD and ConfSolv literature.^{9;24–29} The OpenFF 2.3.0 calculation records and the GBn2 dataset used here are identified separately in Supplementary Methods S3. The Abraham descriptors quantify excess molar refraction (E), dipolarity/polarizability (S), hydrogen-bond acidity and basicity (A, B), and gas-to-hexadecane partitioning (L). Directed message-passing neural networks (D-MPNNs) learn the two continuum-solvent targets; tree models learn the remaining targets. Surrogate architectures and training details appear in Methods.

The target is a molecular equilibrium property, which includes conformational averaging. A single molecule can have many conformers but does not have a separate experimental equilibrium hydration free energy for

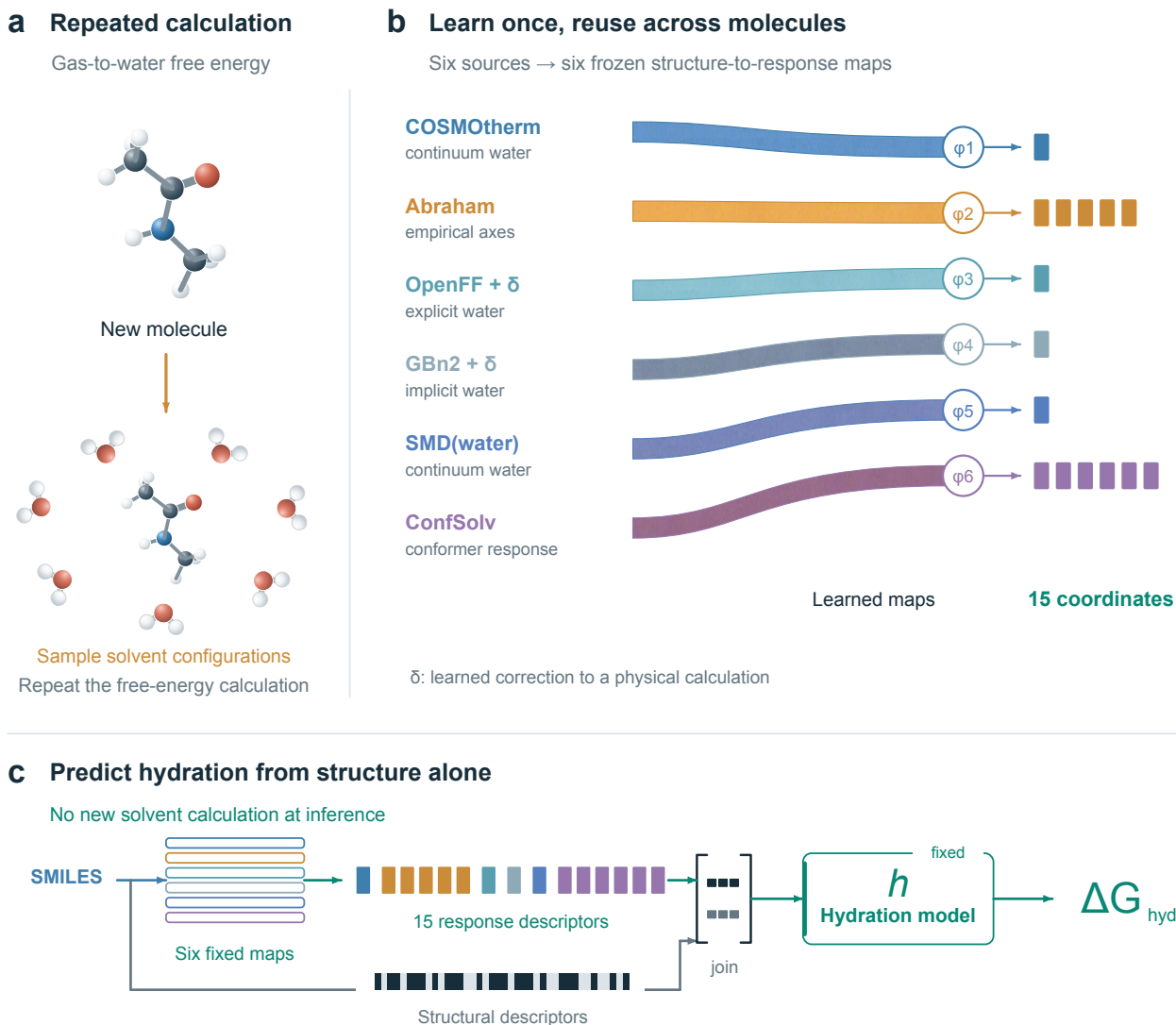


Figure 1: Solvent-response information can be learned once and reused. **a**, Gas-to-water transfer illustrates the physical information required for hydration prediction; explicit sampling incurs a new cost for each molecule. **b**, Six source families supply complementary targets. Separate surrogates learn to predict these quantities from structure and are frozen before endpoint fitting. Colors connect each source to its predicted descriptors; the counts 1 + 5 + 1 + 1 + 1 + 6 sum to 15. OpenFF and GBn2 each combine a physical-energy prediction with a learned experimental residual, δ . A separate hydration predictor combines the predicted responses with structural descriptors and learns from experimental hydration labels. **c**, Deployment evaluates the frozen surrogates and predictor from a SMILES string, without a new solvent calculation. Molecular structures and solvent shells are explanatory illustrations, not simulated configurations. The six colored streams depict separate information sources, not calculated response fields. The hydration mapping h is architecture-neutral; the tested implementations are compared in Table 2.

each one. Our SMILES-based model predicts that molecular property, not the hydration energy of a specified geometry or a conformationally constrained state. The six ConfSolv descriptors provide learned summaries of conformational effects; they do not reconstruct a molecule’s conformer ensemble at inference. This distinction also motivates our comparison with Uni-Mol, a pretrained representation that uses a generated three-dimensional conformer.³⁰ Supplementary Fig. 7 illustrates the conformational targets and their relationship to the molecular equilibrium property.

Results

Separating source supervision from hydration evaluation

We evaluate the predictor in two settings with different roles. ARROW-85 is a small reference set with experimental and high-fidelity simulation values for the same molecules. The external set contains 220 neutral

Table 1: The 15 response descriptors and the information they provide. Every listed quantity is predicted from molecular structure at inference. CombiSolv-QM supplies COSMOtherm targets, SoluteML supplies Abraham parameters, and MolSolv supplies SMD(water) targets. Complete definitions, units and source counts are in Supplementary Table 1.

Source	Number	Predicted quantity	Surrogate
CombiSolv-QM	1	COSMOtherm water transfer free energy	D-MPNN
SoluteML	5	Abraham E , S , A , B , L interaction parameters	ExtraTrees
OpenFF	1	Explicit-solvent estimate plus experimental residual	ExtraTrees
GBn2	1	Implicit-solvent estimate plus experimental residual	ExtraTrees
MolSolv	1	SMD continuum-water transfer free energy	D-MPNN
ConfSolv	6	Conformer corrections and water-response distributions	LightGBM

solutes derived from original Henry-law measurements, enabling a larger test outside the endpoint training set. In both settings we compare predictions against experiment; the published ARROW simulation results are a separate accuracy reference and never a SolvAI input.

The source datasets differ in scale and physical resolution. MolSolv contributed 1,729,545 SMD(water) calculations, of which 350,359 structures were used to train the confirmatory surrogate after quality and identity filtering.³¹ ConfSolv contributed 5,392,567 water conformer records; aggregation, target availability and standardized-identity filtering left 17,829 connectivities for the response model. The archive counts refer to calculations for MolSolv and conformer records for ConfSolv; the retained training rows are molecule-level records. Complete definitions appear in Supplementary Table 1 and source-specific provenance in Supplementary Fig. 2.

The experimental training pool contains 1,280 public hydration labels, supplemented by the available ARROW training folds. Every response-source table excludes exact and standardized equivalents of the 85 ARROW test identities. The 220 external molecules are absent from all endpoint labels; 97 are also absent from every response-source table. We report the full cohort and this nested subset because a known source response and a known hydration label are distinct forms of supervision. The exact exclusions and the three affected surrogate refits are documented in Methods, Supplementary Table 2 and Supplementary Data 4.

One response representation, different endpoint behaviors

The 15 frozen response predictions can be combined with different molecular representations and endpoint learners. We evaluated ExtraTrees, residual multilayer perceptrons (MLPs), separate structure and response branches, a linear predictor with a neural residual, message-passing graphs with size-sensitive readouts, frozen MoLFormer representations with neural heads, TabM parameter-sharing ensembles, and TabPFN-3.5 in-context prediction (Table 2). TabM shares most weights across multiple neural predictions; TabPFN uses a pretrained transformer conditioned on labelled training rows.^{32–34} A MoLFormer–ExtraTrees fit changes the representation while retaining a tree endpoint. Supplementary Figs. 8–10 depict the mechanisms. We ask separately which endpoints predict most accurately and whether adding the responses helps each one; these need not have the same answer.

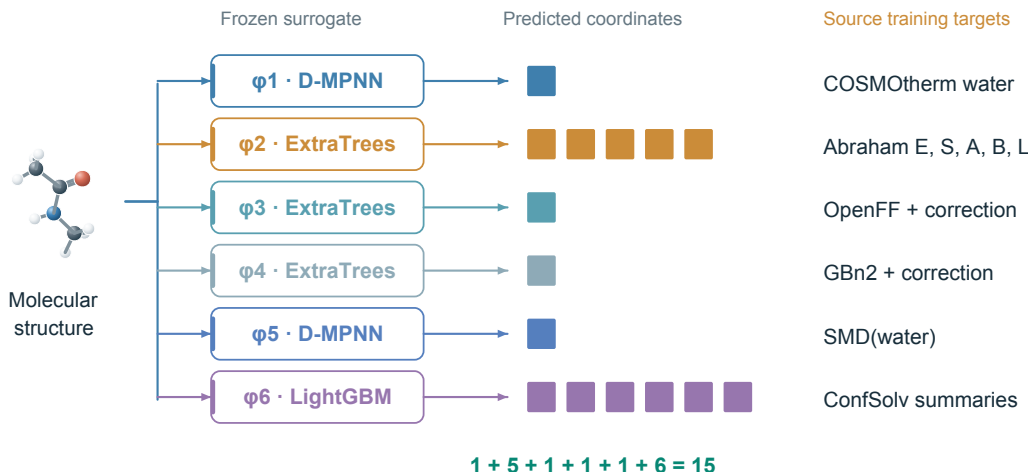
This architecture comparison is retrospective: the evaluation sets were already known during method development. Within each outer partition, configuration and training duration were chosen using only training-side validation data. We completed longer-training, representation and width challenges, locked the selection policy, and then fitted the final models. The policy selected TabPFN in all five ARROW folds and the external fit. Input recipes can differ by partition; the external TabPFN fit uses RDKit descriptors, a frozen MoLFormer encoding and the responses. No-response controls inherit each selected recipe; they are paired ablations, not separately optimized structural baselines. The common comparison for all nine fixed configurations uses the same ARROW-85, External-220 and nested Strict-97 evaluation identities, with a paired no-response fit for each configuration. Eight configurations were also refitted for the separate size-held-out diagnostic; the MoLFormer–ExtraTrees combination was not. Shuffled-response and chemical-separation results below concern the ExtraTrees fits and are not claims about the other endpoints.

The lowest full-model point estimate changes with the evaluation: TabPFN has 0.199 kcal mol^{−1} on ARROW-85, a MoLFormer neural head has 1.111 on External-220, ridge plus a neural residual has 1.429 on Strict-97, and TabPFN has 1.318 in the exploratory size refit (Table 2). ExtraTrees also fits the small reference set closely at 0.202 but is less accurate on the size holdout. These ranks are descriptive, not a statistically established ordering of all endpoint pairs.

Within-family response effects tell a different story (Fig. 3). On External-220 and its nested Strict-97 subset, seven of the nine fixed endpoint configurations have descriptive paired intervals below zero when the responses are added. These include ExtraTrees, the residual MLP, dual branches, message passing, a MoLFormer neural head, TabM, and MoLFormer with a tree head. On ARROW-85, the intervals are below zero for ExtraTrees and

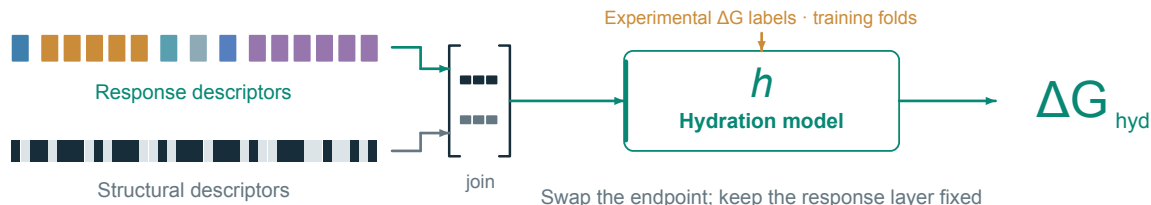
a Distil each source into a structure-to-response map

Source targets are computed quantities or empirical measurements



b Learn the endpoint separately

Experimental hydration labels supervise the endpoint, not its source targets



c Distinguish endpoint exclusion from source exclusion

85 ARROW molecules

Absent from all six supervised sources

Each hydration prediction is out of fold:
its label is excluded from that endpoint fit.

220 external molecules

All absent from endpoint training



123 source-exposed

97 absent from all six

97 is a subset of 220, not a separate fit.

Figure 2: Source supervision and endpoint labels play different roles. **a**, The six source-specific surrogate families comprise two D-MPNN, three ExtraTrees and one LightGBM family, with target-specific fits within the tree-based families, to predict 15 descriptors. The source targets differ; each mapping is frozen before the hydration predictor is fitted. **b**, An endpoint model h combines these predictions with structural features and learns from the available experimental hydration labels. The symbol h denotes this learned mapping without specifying its architecture; Table 2 compares the implementations. **c**, Identity exclusion distinguishes knowledge of a source response from knowledge of the endpoint. All 85 ARROW identities are excluded from the six supervised source tables; each ARROW endpoint prediction is also out of fold. All 220 external molecules are absent from endpoint training. Of these, 97 are additionally absent from every response-source table, whereas 123 occur in at least one. The strict 97 are a nested subset, not an independent cohort or a separately fitted model. Identity checks include standardized equivalents. These exclusions concern supervised sources, not membership in the original pretraining corpora of external encoders.

the two MoLFormer endpoints. The neural MoLFormer fit changes from 0.465 to 0.273 on ARROW, 1.332 to 1.111 externally, and 1.685 to 1.450 on Strict-97. The residual MLP’s external response gain does not compensate for its higher absolute error. Ridge plus a neural residual gives the lowest full-model Strict-97 point estimate, but removing responses lowers its error further, from 1.429 to 1.294. TabPFN has the lowest ARROW point estimate, yet its response-minus-no-response intervals include zero on all three original cohorts.

The mechanisms therefore illuminate different trade-offs rather than a single winner: separate input branches and the shared-parameter TabM ensemble improve the size-holdout estimate; message passing makes molecular size available to its readout without imposing additivity; and the MoLFormer–ExtraTrees pair changes the

Table 2: Endpoint mechanisms and their measured accuracy. All entries include the 15 response descriptors. MAE is in kcal mol^{-1} . The first three columns use the ARROW and external evaluation identities; Strict-97 is nested within External-220. Neural and TabPFN predictions average five seeds, tree controls three. The final column is a separate exploratory size-held-out refit on 1,160 training and 205 held-out molecules, using three seeds for every method. A dash means not evaluated, not zero error. Supplementary Table 13 includes matched no-response controls and common-three-seed results.

Endpoint	Mechanism and observed pattern	ARROW	External	Strict	Size
ExtraTrees	Leaf averages; bounded output	0.202	1.153	1.536	1.772
Residual MLP	Residual links; external response gain	0.340	1.428	2.016	1.813
Dual branch	Separate inputs; improved size transfer	0.288	1.234	1.616	1.560
Ridge + neural residual	Linear trend + correction; low Strict error	0.336	1.172	1.429	1.391
Message passing	Size-aware graph; external response gain	0.288	1.168	1.505	1.729
MoLFormer + neural head	Frozen sequence encoding; low external error	0.273	1.111	1.450	1.609
TabM	Shared neural ensemble; low size error	0.291	1.132	1.454	1.389
TabPFN-3.5	Label-conditioned transformer; low size error	0.199	1.134	1.493	1.318
MoLFormer + ExtraTrees	Frozen encoding + trees; input contrast	0.234	1.174	1.564	—

molecular representation while retaining a bounded endpoint. A width-adaptive selection variant reaches 1.090 externally, but its difference from ExtraTrees is unresolved (-0.062 ; 95% interval, -0.157 to 0.027). This variant is a training-side selection rule that can choose different configurations across partitions, not another architecture. Complete variants, unsuccessful choices and paired intervals appear in Supplementary Tables 13–15 and Data 6.

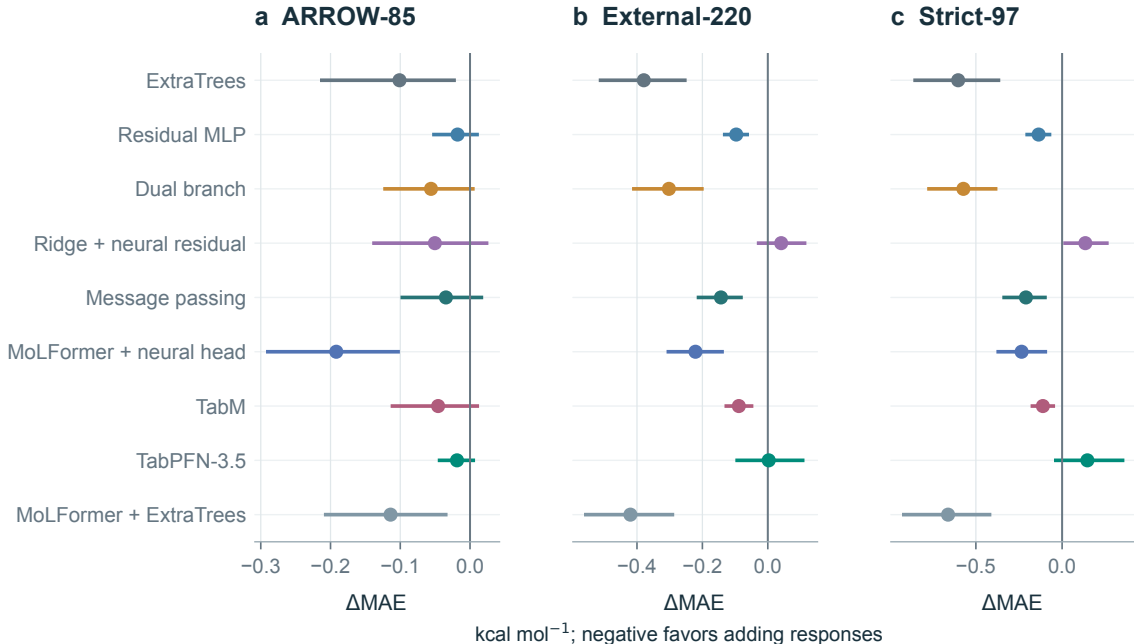


Figure 3: The response contribution varies among endpoint families. a–c, Paired MAE changes from adding the same 15 response descriptors to each selected endpoint recipe on ARROW-85, External-220 and Strict-97. Negative values favor response augmentation. Bars are descriptive 95% percentile intervals from 100,000 paired molecule resamples, conditional on the fitted models and without multiplicity correction. They are not uncertainty intervals for individual molecular predictions. Strict-97 is nested within External-220. The no-response controls inherit the full model’s recipe. Cross-architecture accuracy and seed counts are given in Table 2; family-specific configuration choices are retained in Supplementary Data 6.

Molecule alignment and matched controls

The matched, shuffled, chemical-separation and source-composition experiments below use ExtraTrees endpoints. In five-fold out-of-fold (OOF) prediction, each ARROW molecule is evaluated by a predictor that did not receive its experimental label. Removing only the predicted responses from that predictor gives an MAE of 0.303 kcal mol⁻¹ (Fig. 4b). Adding the eight COSMOtherm, Abraham and corrected-solvent descriptors lowers the error to 0.212; adding SMD(water) gives 0.207; and adding the six conformer descriptors reaches 0.202 kcal mol⁻¹. The last value lies on the same accuracy scale as ARROW/PIMD8 (0.205 kcal mol⁻¹), although the paired comparison does not establish superiority over PIMD8.

At the molecule level, the response-augmented ExtraTrees endpoint reduces the absolute error for 53 of 85 solutes and reduces MAE by 0.101 kcal mol⁻¹ relative to the matched endpoint (95% paired-bootstrap interval for responses minus the matched endpoint, -0.215 to -0.020; Fig. 4c). The empirical and residual-corrected subset also reduces MAE, with a paired interval entirely below zero. The three computed-energy descriptors (COSMOtherm and uncorrected OpenFF/GBn2), SMD alone and ConfSolv alone yield smaller MAE reductions in isolation than the complete response set, whose paired interval lies entirely below zero (Fig. 4d). The comparison measures the predictive benefit of combining these descriptors. Molecule-level prediction, residual and paired-change diagnostics are shown in Supplementary Fig. 1 and Supplementary Data 2.

Shuffling tests whether the benefit depends on assigning the responses to the correct molecule (Fig. 4a). Across five independent permutations of the 15-column block, the fixed-partition mean is 0.306 kcal mol⁻¹, close to the 0.303 matched control and far from the aligned-response fit. The paired difference between aligned and mean-shuffled descriptors is -0.104 kcal mol⁻¹ (95% interval, -0.214 to -0.025). Alignment between a molecule and its predicted response is therefore necessary for the observed gain.

Partition choice changes the absolute point estimate but not the improvement over the matched control. Across five complete five-fold partitions, matched structure-only prediction gives 0.313 kcal mol⁻¹ on average, whereas the response-augmented fit gives 0.207 ± 0.004 kcal mol⁻¹; the response-augmented endpoint is better in all five partitions (Fig. 4e; Supplementary Table 4). This places the observed error near the PIMD8 reference on this chemistry across partitions, without establishing superiority over that simulation method. Changing the prospectively frozen weight of available ARROW outer-training labels from 3 to 1 leaves the contrast essentially unchanged: 0.310 for structure only and 0.206 kcal mol⁻¹ with responses (paired change -0.103 kcal mol⁻¹; 95% interval, -0.219 to -0.020; Supplementary Table 9).

Responses alongside pretrained molecular features

An additional fixed-protocol comparison isolates the contribution of responses beside pretrained molecular features (Fig. 5). It includes a D-MPNN trained from random weights, a D-MPNN fine-tuned from pretrained CheMeleon weights, and frozen molecular encodings from MolFormer (SMILES sequences) and Uni-Mol (three-dimensional conformers). For each frozen encoding, we repeat the same ExtraTrees downstream fit with and without the 15 responses. The direct graph fits are separate baselines, not the paired ablation for those encoders. All models use the same eligible experimental labels and test identities; model-specific training and inner-validation procedures are in Methods.

On ARROW-85, response augmentation lowers the MolFormer-based predictor’s MAE from 0.348 to 0.234 kcal mol⁻¹ and the Uni-Mol-based predictor’s error from 0.334 to 0.220 kcal mol⁻¹. The advantage also persists on the external cohort and its strict source-disjoint subset (Fig. 5c). Each paired interval for these two augmentation contrasts lies below zero. Thus, the response descriptors remain useful alongside both sequence-based and three-dimensional pretrained features, rather than helping only an unaugmented structural model.

The ExtraTrees endpoint with responses has lower error than either directly trained graph model on ARROW-85 and the 220-molecule external cohort. On the strict 97-molecule subset, the difference from the scratch D-MPNN is unresolved: the paired MAE change is -0.023 kcal mol⁻¹ (95% interval, -0.269 to 0.231). These cross-architecture comparisons evaluate the specified training configurations; they do not determine the best achievable performance of each architecture.

Transfer beyond nearby labels

Random molecular folds can leave close chemical neighbors in the public training pool. We therefore applied each separation rule to all endpoint-supervised molecules, not only the 85 reference solutes. For each outer test partition, public and ARROW training rows in the held-out family, scaffold or molecular cluster were removed. A complementary analysis excluded every training row at or above a prescribed Morgan similarity to any test molecule. The response surrogates remain fixed in these tests.

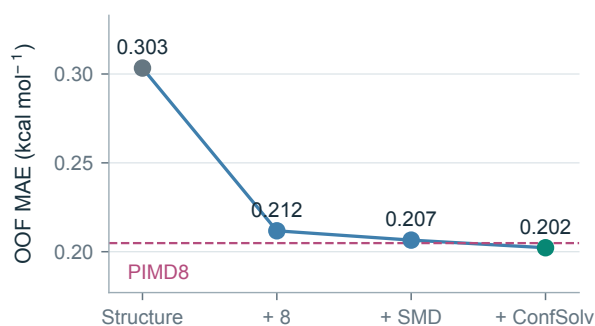
Response descriptors reduce MAE under all predeclared separation regimes (Fig. 6; Supplementary Table 5). At a train-test similarity exclusion threshold of 0.70, MAE falls from 0.343 to 0.211 kcal mol⁻¹; cluster separation lowers it from 0.349 to 0.216. Scaffold and functional-family separation are much harder, but the same comparison

a Change only the response information

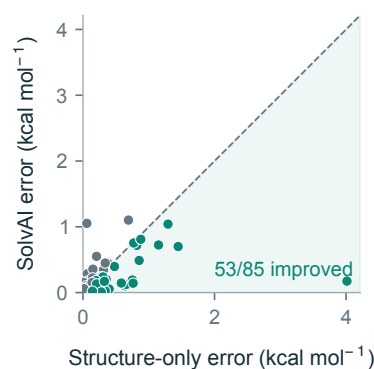
Same structures, labels, folds, weights, learner and seeds



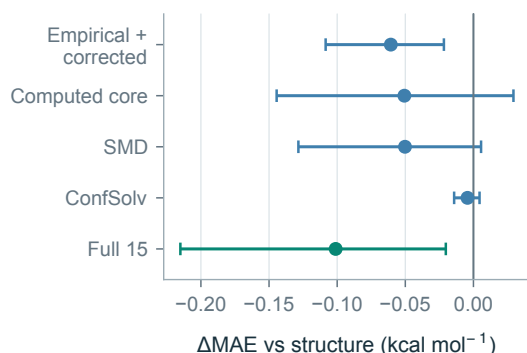
b Build the response set



c Compare individual molecules



d Which source blocks contribute?



e Repeat the full partition

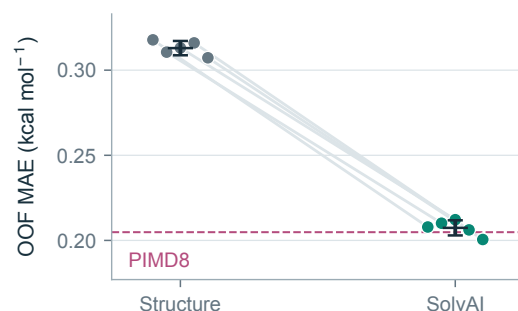


Figure 4: Molecule-aligned response descriptors improve a matched hydration endpoint. **a**, The matched control removes the responses; the shuffled control reassigns the complete response block among molecules. Labels, structural descriptors, architecture, weights, folds and seeds are fixed. The assignment sketches are illustrative; shuffled MAE is the mean of five permutations. **b**, Fixed-partition progression adds eight COSMOtherm, Abraham and corrected-solvent descriptors, then SMD(water), then six ConfSolv summaries. The dashed magenta line marks ARROW/PIMD8 on the same 85 solutes, as an accuracy reference rather than an input or training source. **c**, Paired absolute errors in kcal mol^{-1} for every molecule. Shading marks lower error with responses. **d**, MAE change for predeclared response blocks relative to structure only; bars are 95% molecule-bootstrap intervals from 100,000 resamples. **e**, Paired results for five complete split repeats. Thin lines join methods within the same partition; black marks show mean $\pm$ sample s.d.

a Test alternative molecular representations

Direct graph learning



D-MPNN / CheMeleon

Fit the graph model end to end

Frozen encoder + matched ExtraTrees



b Compare every model on the same held-out molecules



c Add responses to each fixed representation

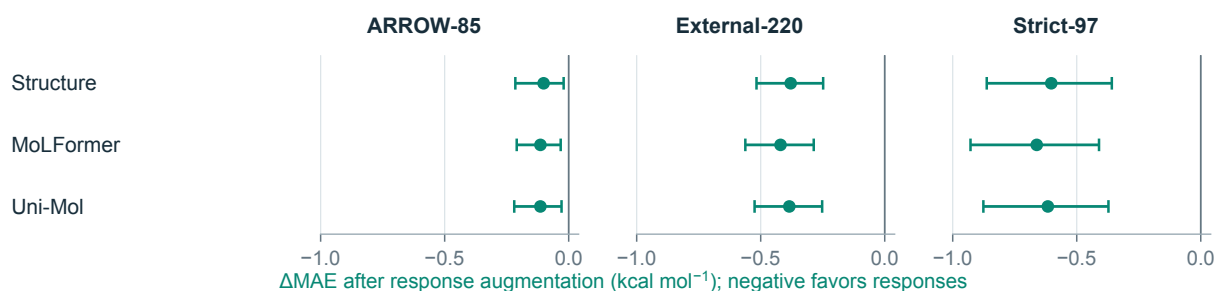


Figure 5: Response descriptors complement pretrained molecular representations. **a**, Direct graph models fit the entire network. Frozen MoLFormer and Uni-Mol representations instead augment the same structural features and ExtraTrees endpoint; paired fits add or remove only the 15 responses. **b**, All nine model conditions on ARROW-85, External-220 and Strict-97. Values are MAE in kcal mol⁻¹; note the different ARROW and external horizontal scales. ARROW predictions are out of fold. External predictions use all 1,365 endpoint-training labels; Strict-97 is a subset of External-220, not a separate fit. The ten-response control removes the Abraham parameters and experimental residual corrections (Fig. 7a). **c**, Paired changes after response augmentation, relative to each unaugmented representation. Negative values favor adding responses; bars are 95% intervals from 100,000 paired molecule resamples. Full numerical results and intervals are retained in Supplementary Tables 11–12.

favors the response-augmented endpoint: 0.728 to 0.376 and 1.262 to 0.468, respectively. Every paired 95% bootstrap interval lies below zero. Response descriptors therefore retain value away from nearby endpoint labels, even where absolute endpoint accuracy is not yet sufficient. Molecule-level family errors and the four global separation regimes are expanded in Supplementary Fig. 5 and Table 6.

We next removed all 85 ARROW experimental labels from endpoint training. Using only the same 1,280 public training labels, the matched structure model gives $0.385 \text{ kcal mol}^{-1}$ and the response-augmented endpoint gives $0.257 \text{ kcal mol}^{-1}$ (paired change -0.128 ; 95% interval, -0.246 to -0.037). This transfer test is not an independent external benchmark – the method was developed in the context of ARROW chemistry – but it shows that response descriptors retain value without fold-local adaptation to the reference set.

Finally, we evaluated a prospectively frozen external cohort derived from original measurements in Sander’s Henry-law compilation.³⁵ Eligibility was determined without reference to model errors. Of 221 pre-existing candidates, one was excluded because an independent registry audit had already established a conflict between the name, Chemical Abstracts Service (CAS) registry number and structure. The remaining 220 neutral solutes have no exact or standardized-equivalent identity in any endpoint label used to fit the deployment models. On this endpoint-disjoint cohort, MAE falls from 1.532 to $1.153 \text{ kcal mol}^{-1}$ (paired change -0.379 ; 95% interval, -0.517 to -0.248 ; Fig. 6d; Supplementary Table 10). Because some molecules occur in response-source tables, we also froze a stricter subset of 97 molecules absent from all six response sources. The advantage remains: 2.138 to $1.536 \text{ kcal mol}^{-1}$ (paired change -0.603 ; 95% interval, -0.863 to -0.359). Thus, the response descriptors remain beneficial beyond ARROW-85, although absolute errors are substantially larger on the external set.

The external cohort also permits a direct view of error against distance from endpoint-training chemistry (Fig. 6e). At nearest-training Morgan similarity below 0.30, 35 molecules have MAEs of $4.027 \text{ kcal mol}^{-1}$ for structural features and 2.501 with responses. The corresponding errors are 1.227 and 1.014 for 103 molecules in $[0.30, 0.50)$, and 0.822 and 0.724 for 74 molecules in $[0.50, 0.70)$. The eight molecules at similarity 0.70 or higher give an imprecise contrast, with a paired interval spanning zero. Thus, the largest absolute improvement occurs in the least similar bin, where substantial residual error also remains.

Source composition changes transfer behavior

The response set mixes computed quantities with empirical parameters and experimental residual corrections. To distinguish their roles, an additional control (Fig. 7) retains ten computed-source descriptors: COSMOtherm, uncorrected OpenFF and GBn2, SMD(water), and the six ConfSolv summaries. The predictor, experimental training labels and weights remain unchanged. On ARROW-85 this control reaches $0.234 \text{ kcal mol}^{-1}$; its improvement over structural features alone has a paired interval spanning zero (-0.191 to $0.022 \text{ kcal mol}^{-1}$). The full 15-descriptor model is better than this control by $0.032 \text{ kcal mol}^{-1}$ (95% interval for full minus computation-only, -0.063 to -0.006).

The ordering reverses outside the reference chemistry. The ten-descriptor model reaches $1.095 \text{ kcal mol}^{-1}$ on the external 220 molecules, compared with 1.153 for the full model. The paired full-minus-computation-only difference is $0.057 \text{ kcal mol}^{-1}$ (95% interval, 0.023 to 0.093). On the strict 97-molecule subset, the corresponding errors are 1.422 and $1.536 \text{ kcal mol}^{-1}$, with a paired interval of 0.058 to 0.172 for the difference. Computed-source descriptors therefore reduce external error relative to structural features alone, while the full source composition is more effective on ARROW. This comparison identifies a dependence on source composition and evaluation chemistry, rather than a single source set that is uniformly preferable. Both fixed configurations are reported; external errors were not used to retune either model.

When response prediction is insufficient

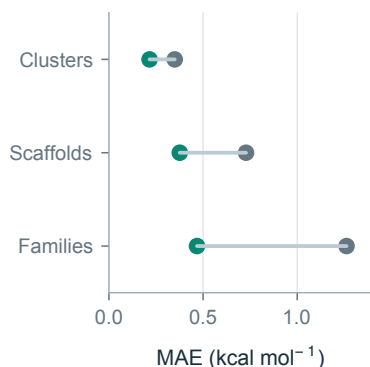
The choice of response target affects whether its predicted value improves hydration. In exploratory development screens, compact ConfSolv response summaries transferred better than graph or feed-forward latent representations trained on the same source. Other candidate signals – including additional force-field diagnostics, pair-interaction data and learned implicit-solvent hierarchies – were non-improving in matched exploratory screens (Supplementary Fig. 3). These screens test whether each representation can be inferred from structure accurately enough, and with sufficiently non-redundant information, to improve the endpoint.

We also attempted to predict the mean derivative of interaction energy along an alchemical coupling path, separately for the total, electrostatic and van der Waals components. Source targets came from short two-bead path-integral simulations (PIMD2) at coupling values $\lambda = 0.1, 0.5$ and 0.9 . Structure-only predictors captured broad response trends, but component MAEs ranged from 1.27 to $5.20 \text{ kcal mol}^{-1}$ (Supplementary Fig. 4). Adding the predicted PIMD2 response worsened the matched exploratory endpoint from 0.196 to $0.201 \text{ kcal mol}^{-1}$; a predicted classical-NQE-PIMD hierarchy reached 0.215 , and their combination 0.219 in that screen. Direct integration of the predicted response curve, with fold-local affine calibration, yielded $1.51 \text{ kcal mol}^{-1}$ (Supplementary Fig. 4). No PIMD-derived coordinate is present in the final model.

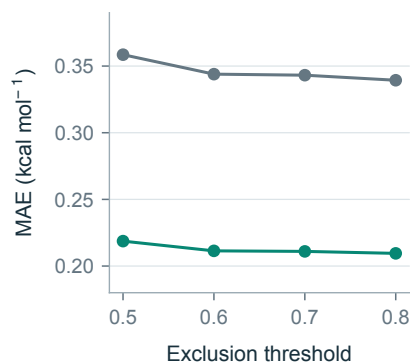
These exploratory results suggest three considerations when choosing response targets: relevance to hydration,

Structure only ● SolvAI ●

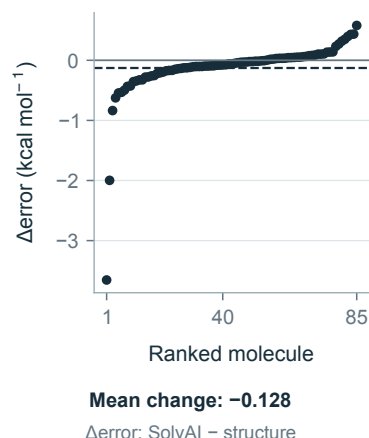
a Chemical groups



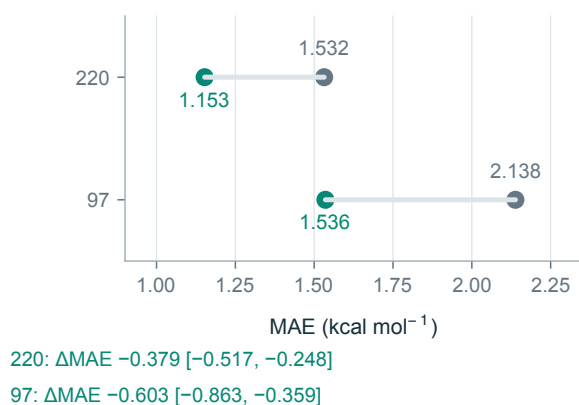
b Nearest neighbors



c No ARROW labels



d External matched comparison



e Relate error to chemical distance

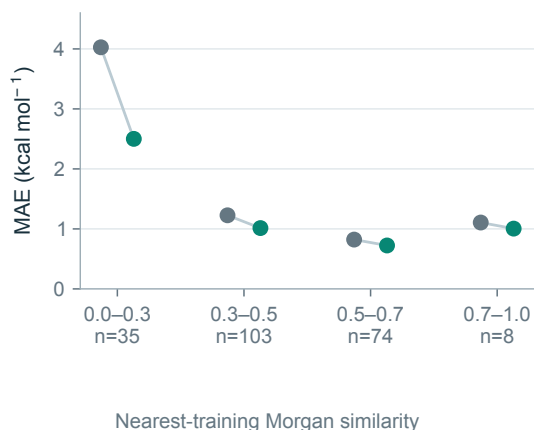
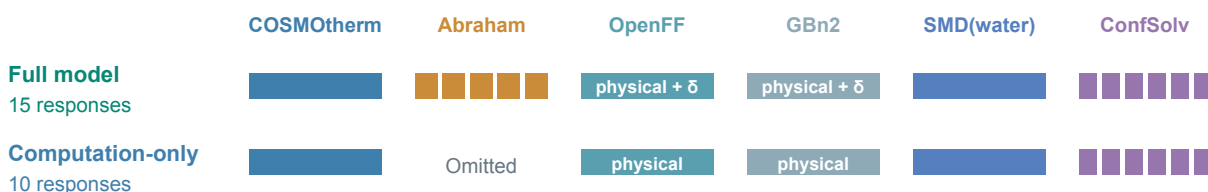


Figure 6: Response descriptors reduce error under chemical separation. **a**, Matched MAEs after removing every endpoint training label in the held-out molecular cluster, scaffold or functional family. **b**, Similarity exclusion removes every training row at or above the indicated Morgan similarity threshold to a test molecule. **c**, Molecule-level change in absolute error when no ARROW experimental label is used for training; negative values favor adding responses. **d**, Matched performance on the prospectively frozen external cohort and the subset also absent from every response-source table. Bracketed values below the plot are 95% intervals from 100,000 paired molecule resamples for the MAE difference, not intervals for the plotted MAEs. **e**, External errors in fixed nearest-training Morgan similarity bins, with all sample counts. The largest improvement occurs in the least similar bin, where absolute errors also remain largest. These bins describe the cohort, not calibrated applicability thresholds. All response surrogates use the standardized-equivalent exclusions described in Methods.

a Keep the learner fixed, change the response sources



Remove five Abraham values and two residual corrections (δ). Retain the two physical energies.

Both models learn hydration from the same experimental labels.

b Compare absolute errors



c Compare the source sets directly

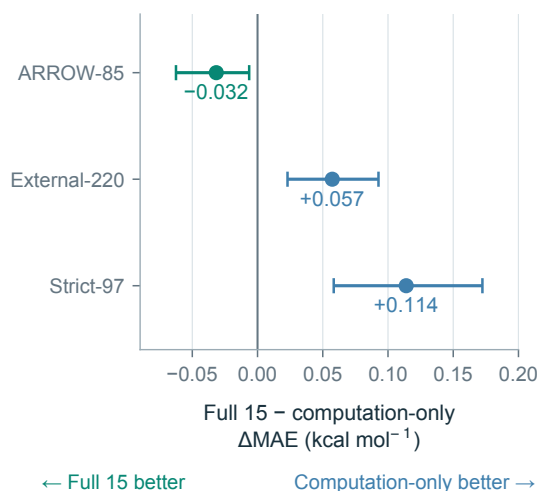


Figure 7: The preferred source composition changes with evaluation chemistry. **a**, The computation-only control removes five Abraham descriptors and both experimental residual corrections, δ , while retaining the underlying OpenFF and GBn2 physical-energy predictions. The resulting computation-only set contains ten descriptors. Both models retain the same structural features, experimental endpoint labels, weights and learner. Each tile denotes one descriptor, not its magnitude. **b**, Absolute MAEs for structural features, all 15 responses and the ten-response control on all three reported evaluation sets. **c**, Paired full-minus-computation-only MAE differences with 95% intervals from 100,000 molecule resamples. Negative values favor the full model on ARROW-85; positive values favor computation-only responses externally. The strict 97 are nested within the external 220. The external extension is retrospective and compares two fixed configurations without retuning on external errors. The reversal identifies a dependence on source composition and chemistry, not one uniformly optimal response set.

accuracy of their structure-based prediction, and complementarity to the other features. They do not isolate a causal effect of physical fidelity, because the source sizes, targets and learned representations also differ. The exploratory analyses are reported separately in Supplementary Figs. 3–4 and Supplementary Data 1.

Testing transfer to larger molecules

To test accuracy beyond the endpoint’s training-size range, we held out the largest 15% of molecules within each endpoint source and refitted eight of the nine fixed endpoint configurations on the remaining 1,160 labels. The MoLFormer–ExtraTrees combination was not included in this diagnostic. The 205 held-out molecules contain 8–37 heavy atoms; 188 exceed the largest training molecule, which has 17. The overlapping lower end results from applying the rule within each source. These labels were available during earlier random-fold development, so this is an exploratory size diagnostic, not an untouched prospective test. The response surrogates remain fixed: this comparison tests endpoint transfer conditional on their predictions, not size extrapolation with all response-source exposure removed.

TabPFN reduces the size-held-out MAE from 1.772 for ExtraTrees to 1.318 kcal mol^{−1} (paired difference −0.453; 95% interval, −0.648 to −0.281; Fig. 8a). Here its response descriptors also help: without them the error is 1.959 (response-minus-control difference −0.640; interval, −0.847 to −0.451). The refits use separately locked input recipes, so the cross-endpoint contrast does not isolate architecture from representation or preprocessing. TabM, the linear-plus-neural model and the dual-branch model also reduce error relative to ExtraTrees, with paired intervals below zero. These results identify useful alternatives for this size split; they do not imply the same ordering for every new chemical family (Supplementary Table 15; Supplementary Data 6, `size_diagnostic_comparisons.csv`).

Capped glycine and alanine series ask a complementary question about prediction shape. ExtraTrees approaches plateaus near −23 and −25 kcal mol^{−1}, whereas several neural endpoints and TabPFN continue to change over the tested range (Fig. 8b,c). For Ac-(Gly)_n-NHMe, TabPFN predicts −40.802 at $n = 6$ and −67.194 at $n = 12$, compared with −23.005 and −23.440 for ExtraTrees. No experimental peptide values are available here: these are predictions, not errors, and a more negative or more linear curve is not evidence of greater physical accuracy. The finite outer bound on the mean prediction is approximately [−32.595, 5.757] for the released trees and [−658.495, 649.922] for the fitted TabPFN ensemble. The latter is much wider but still finite; neither architecture guarantees asymptotic extensivity.

Linear alkanes expose an additional limitation. TabPFN remains non-monotonic between C10 and C14 (3.348 and 3.046 kcal mol^{−1}), despite its changed peptide behavior. Moreover, 27 of the 30 tested alkanes occur in endpoint training, so this series is not an independent extrapolation benchmark. A source audit confirmed that the low C14 and C16 training values were copied faithfully from their source; removing these two labels raises the corresponding tree predictions. We retain the original labels and report the sensitivity rather than substitute an assumed linear trend (Supplementary Methods S20).

Finally, the native TabPFN predictive distributions do not yet provide calibrated reliability. Equal mixtures of the five fitted distributions give nominal 80% and 95% intervals with only 58.2% and 80.0% empirical coverage on External-220; the corresponding Strict-97 coverages are 55.7% and 76.3%. These are predictive intervals, not the spread across seeds. No calibration or model reselection was performed on these outcomes (Supplementary Fig. 11 and Table 16).

Discussion

SolvAI’s response representation can be reused across endpoint architectures, but its contribution depends on the learner and evaluation set. Separate surrogates learn the source quantities, and an experimentally supervised predictor learns how their predicted values combine for hydration. Matched comparisons show response benefits for several endpoints on at least one evaluated cohort, including ExtraTrees, a MoLFormer neural head, message passing and TabM. TabPFN is competitive on the reference set without a resolved response increment there or on the original external cohort. No endpoint has the lowest error in every evaluation; endpoint and input-recipe choice change the balance among small-molecule accuracy, external transfer and size transfer.

Within ExtraTrees, molecule shuffling shows that the observed benefit requires correct alignment between molecule and response prediction. Chemical separation and the external cohort show where this benefit persists as the relationship to training chemistry changes. These controlled results establish a feature contribution for that learner; they do not establish the same effect size for every endpoint.

A descriptor set that is best on reference chemistry can transfer less well than a smaller calculated-source set. Adding empirical response information and residual corrections therefore has to be assessed in the intended domain, even when the information improves an in-domain benchmark. Our comparison establishes this reversal for the two evaluated cohorts; choosing sources for a new deployment domain would require validation data from that domain. The original 15-descriptor deployment model remains unchanged.

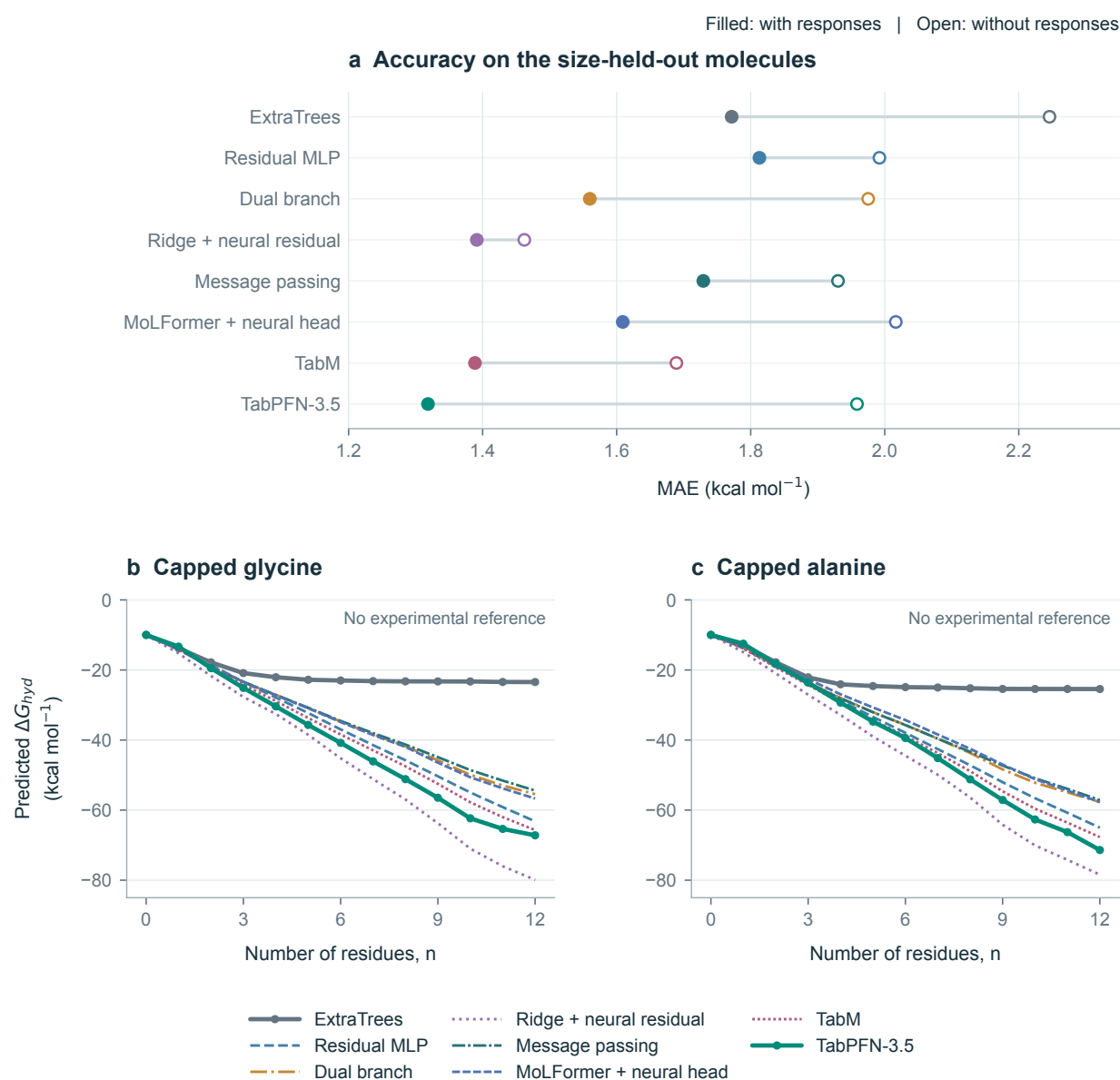


Figure 8: Labelled size transfer and unlabelled series answer different questions. **a**, MAE on the 205-molecule size holdout after refitting on the same 1,160 training labels. Filled and open marks denote models with and without responses. All predictions average three seeds. This diagnostic reuses development labels under a different split. **b,c**, Predictions along capped glycine and alanine series, using the full-data models with responses. Neural and TabPFN curves average five seeds; ExtraTrees averages three. These curves have no peptide reference values and therefore do not rank physical accuracy. The abscissa is the number of residues between the acetyl and N-methylamide caps; $n = 0$ is N-methylacetamide. Colors identify the same families in both panels.

The size-held-out diagnostic makes this endpoint-and-input dependence concrete. The selected TabPFN recipe lowers error relative to the selected ExtraTrees recipe and gains from the response descriptors on that split, although its original external-cohort response increment is unresolved. Thus the same representation is not a guarantee of improvement for every learner or data regime. The response sources were kept fixed rather than changed to suit the outcome of each evaluation.

Architectural flexibility and physical transferability are different properties. Removing the narrow tree-output range changes the peptide-series predictions, but does not establish the correct free-energy increment per residue. Neither a neural head nor in-context prediction supplies conformer-resolved energies, forces, new-solvent dependence or exact size additivity without corresponding structure and supervision. The size-held-out labels provide a more direct accuracy test than unlabelled peptide curves, but remain an exploratory reuse of the development pool.

Predicted physical descriptors, Δ -learning and computed-to-experimental transfer are important precedents for this design.^{8;9;18;21} The contribution here is their application to complementary solvent-response targets with explicit tests of what those targets add. In these comparisons, the response descriptors can be removed, shuffled, or added to a different representation without changing the experimental target.

The transfer criteria are also consistent with systematic studies of quantum descriptors, which emphasize endpoint relevance and accurate inference.¹⁹ The present controls also test complementarity to the deployed molecular representation. Recent work found that hidden surrogate representations can outperform explicit predicted descriptors in some small-data tasks.²⁰ Here, by contrast, compact ConfSolv response summaries outperformed graph and feed-forward latent coordinates from the same source. The ordering is task-dependent: the useful representation depends on which physical information survives structure prediction and remains complementary to the endpoint features. The transferable signal is therefore not "simulation" in the abstract, but an aligned, learnable representation of solvent response.

On the release host, the packaged ExtraTrees implementation’s warm CPU median is 15.29 s for one molecule and 15.82 s for a batch of 32 (0.494 s per molecule). Its computational advantage is the removal of per-query physical simulation, a property shared with the direct molecular-learning baselines. The added role of SolvAI is to reuse source-response supervision through explicit learned descriptors. Comparisons with the published 15-window PIMD8 workload concern total work on different hardware, not a same-hardware kernel benchmark.

Validation currently covers neutral small molecules near room temperature, with method development centered on the 85-solute reference set. The five-partition MAE of SolvAI-ExtraTrees is 0.207 kcal mol⁻¹, increasing to 0.376 and 0.468 under scaffold and functional-family separation. On the external cohort and strict subset, the full ExtraTrees model retains a statistically supported reduction in error relative to structural descriptors, but has larger errors of 1.153 and 1.536 kcal mol⁻¹. These errors, and their dependence on chemical similarity, motivate domain-specific validation before deployment. Per-query reliability remains uncalibrated: the nominal 95% TabPFN intervals cover only 80.0% of the external cohort. Ions, salts, biomacromolecules and other solvents remain unvalidated; the unlabelled size series do not expand that validated domain.

The unsuccessful alchemical-response experiment identifies a separate data need. Predicting the interaction-energy derivative along a coupling path well enough to integrate it requires more accurate component curves than the sparse PIMD2 sources provided here. Additional protocol-matched response data would test that route; adding more experimental endpoint labels alone would not supply those missing curves.

The broader opportunity is to treat costly computation as scientific supervision with a lifetime beyond one query. Once a useful response is represented across enough chemistry, its information can be predicted from structure and reused without another solvent simulation. Hydration provides a demanding demonstration because the accuracy reference includes explicit solvent and nuclear quantum effects. The same principle may apply wherever physical computation exposes transferable response descriptors that a deployed model can learn.

Methods

Reference set and targets

The primary evaluation uses the 85 neutral organic solutes reported in the ARROW hydration analysis at 298 K.⁵ One water condition was retained per connectivity. The released benchmark table contains canonical structure, experimental hydration free energy, experimental uncertainty where available, classical ARROW, four-bead PIMD (PIMD4) and PIMD8 values, and all fold assignments. The exact reconstructed PIMD8 MAE is 0.205 kcal mol⁻¹; the source article reports approximately 0.20 kcal mol⁻¹.

Endpoint labels and features

The public training pool contains 1,280 neutral, benchmark-disjoint experimental hydration labels: 1,147 connectivity-resolved records from a FreeSolv/CombiSolv-Exp merge and 133 SoluteML dGsolvDB3 records supported by at least two source measurements.^{2;9;25} The ExtraTrees input has a 2,048-bit radius-2 Morgan

fingerprint and 217 RDKit two-dimensional descriptors. Non-finite values are handled by a median imputer fitted inside each endpoint fit.

Response surrogates

Fifteen response descriptors are generated by six surrogate families (Supplementary Table 1). CheMeleon-initialized D-MPNNs predict CombiSolv-QM/COSMOtherm and MolSolv SMD(water) targets.^{36;37} ExtraTrees models predict five Abraham axes and the physical and residual components of corrected OpenFF and GBN2 responses. LightGBM models predict six ConfSolv conformer and response-distribution summaries. The latter are derived from the H₂O subset of ConfSolv. The final coordinate list, target units, source sizes, transformations and inference calculations are given in Supplementary Table 1. PIMD-derived targets are absent from this list.

For the ExtraTrees endpoint, an ensemble of three `ExtraTreesRegressor` pipelines²³ is used, each with 360 trees, maximum feature fraction 0.7, minimum leaf size 2, no bootstrap and random seeds 11, 29 and 47. Predictions are averaged. Public endpoint rows have sample weight 1 and available ARROW outer-training rows weight 3. The matched structure-only model is identical except that the 15 response columns are removed.

Identity standardization

RDKit 2026.03.5 was used to generate canonical isomeric SMILES, full InChIKeys and connectivity blocks.³⁸ The confirmatory audit additionally applied MolStandardize cleanup, largest-fragment selection, uncharging and canonical tautomer enumeration. Exact and standardized benchmark equivalents were removed from every supervised external source. The audit found no such match in the endpoint pool. It identified 2 CombiSolv-QM, 32 MolSolv and 22 ConfSolv source records beyond the original exact-connectivity exclusion. The affected surrogates were refitted with all remaining molecules held in their original train, validation or test split. Full pair lists, identifiers and split-verification hashes are released.

Confirmatory evaluation

Confirmatory analyses followed a prespecified protocol archived in Git. Documented implementation amendments preceded the corrected runs (Supplementary Methods S9). The primary partition is the existing five-fold assignment on the 85 molecules. Five complete shuffled K-fold partitions use seeds 314159, 271828, 161803, 141421 and 173205. In each outer fold, all 1,280 public labels and the available ARROW outer-training labels are fitted; the test target is absent. Feature blocks, ExtraTrees hyperparameters, sample weights and model seeds are identical across all comparisons.

A separately prespecified one-factor sensitivity repeated the fixed folds after changing only the available ARROW outer-training weight from 3 to 1. It retained all source rows, surrogates, features, hyperparameters and seeds. Its paired interval used the same 100,000-resample convention.

For the negative control, the 15 response columns are permuted as one block, thereby preserving their joint distribution while destroying molecule alignment. Public, outer-training and outer-test response rows are permuted separately within each fold. Five permutation seeds (88001–88005) are evaluated for the fixed partition and each of the five repeats.

Chemical separation

Separation is applied to the complete endpoint-supervised pool. Functional families are assigned by a deterministic SMARTS precedence; five outer folds are formed by group. Bemis–Murcko scaffolds are canonicalized, with acyclic structures grouped by primary family. Taylor–Butina clusters use radius-2, 2,048-bit Morgan fingerprints at Tanimoto similarity 0.70. For each outer test fold, every public or ARROW training row sharing a held-out group is removed. A separate nearest-neighbor analysis removes every training row with similarity at or above 0.50, 0.60, 0.70 or 0.80 to any test molecule. No threshold was selected by performance. Exact SMARTS, assignments, training counts and maximum retained similarities are released in Supplementary Data 3 and the accompanying `results/confirmatory/` tables.

Zero-ARROW transfer

The frozen endpoint configuration was fitted once to the 1,280 public experimental labels without any ARROW experimental value. Both matched structure-only and full response-descriptor models then predicted all 85 ARROW molecules. This analysis measures transfer without reference-set endpoint adaptation; it is not described as an independent external test because model development used the reference chemistry.

External validation

The candidate cohort was frozen for a separate solvation-model validation study, before that study’s model predictions, using original-measurement records from Sander’s Henry-law compilation v5.0.0.³⁵ The intrinsic Henry solubility constant H_s^{cp} is the aqueous concentration divided by the gas partial pressure. Values at 298.15 K were converted using $K_c = H_s^{cp}RT$ and $\Delta G_{\text{hyd}} = -RT \ln K_c$, corresponding to 1 M ideal gas and 1 M ideal-dilute aqueous standard states. Here K_c is the dimensionless aqueous-to-gas concentration ratio, R the gas constant and T the temperature. Qualification preceded SolvAI evaluation and required one connected neutral solute, supported elements, compatible target metadata and no exact, fragment-parent, uncharged-parent or canonical-tautomer equivalent in the 1,280 public or 85 ARROW endpoint labels. One previously documented registry conflict was excluded, leaving 220 molecules. A strict 97-molecule subset additionally excludes any identity-equivalent occurrence in all six supervised response-source tables.

For the matched ExtraTrees comparison, deployment-style fitting used the same 1,280 public plus all 85 ARROW labels, weights, preprocessing, architecture and seeds; only the 15 response columns differ. External-cohort labels were not used for fitting, surrogate or feature selection, thresholding or model choice in that comparison. The subsequent architecture study reuses this already examined cohort and is therefore retrospective, even though its final models do not fit these labels. The primary paired analysis and bootstrap convention match the confirmatory analysis. Complete eligibility decisions, source-exposure matches and molecule-level predictions are released as Supplementary Data 5. Because the compilation is external and molecular identities are disjoint, we call this an external molecule-disjoint cohort; we do not claim that all historical measurement provenance is independent.

Additional molecular-model comparisons

The graph and pretrained-representation comparisons followed a fixed protocol specified before their fits. These are retrospective extensions on the existing evaluation sets, not a second prospective validation. The same public labels, ARROW folds and sample weights are used for every model. Two direct predictors map molecular graphs to experimental hydration free energy: a D-MPNN trained from random initialization and a CheMeleon-initialized D-MPNN with its encoder fine-tuned.^{36;37} For each outer training set and seed, 15% of each label source is held out internally to select the epoch count by validation MAE, with an 80-epoch cap and patience 12. The model is then initialized afresh from the same random or pretrained starting weights and trained on the complete outer training set for that count. Only this refitted model predicts the outer test set. Three seeds (11, 29, 47) are averaged. Supplementary Fig. 6 illustrates the selection and refit protocol; Supplementary Methods S14 gives the complete configurations.

Two further comparisons use frozen pretrained representations: MolFormer encodes a SMILES sequence, and Uni-Mol encodes an RDKit-generated conformer.^{30;39} Each is concatenated with the 2,265 structural features and fitted with the same ExtraTrees ensemble as the matched control. We repeat the fit after adding the 15 response descriptors. This paired design tests whether the responses add value to pretrained features without changing the downstream learner. The pretrained encoders receive no hydration labels during these fits; their original pretraining membership has not been audited for molecular disjointness. All model families and both members of each paired comparison are reported, irrespective of their ranking.

These matched evaluations compare alternative molecular-learning approaches without conflating different data splits or training resources.⁴⁰ We do not compare the reported errors directly with unfiltered FreeSolv leaderboard values: 80 ARROW identities occur in FreeSolv, and the supervised response sources require their own exclusions. A conventional FreeSolv benchmark would require fold-specific refitting of every affected response source. The external 97-molecule subset instead tests all six response sources with explicitly audited exclusions.

The computation-only source control was specified separately before its ARROW fits. After that comparison, we extended the same ten-descriptor configuration to the external cohort without retuning. Its OpenFF and GBn2 components were recovered from the packaged surrogates and checked against the frozen corrected-response sums before evaluation (Supplementary Methods S17). This sequence distinguishes the retrospective source analysis from the original prospective external evaluation.

Broader endpoint search and numerical reproducibility

The extended study retains the five original ARROW outer folds, the external 220 identities, the strict subset and the 15 frozen response predictions. For each outer training pool, random inner splits hold out 15% of each source. The selection score is the equally weighted mean of public-source and ARROW-source validation MAE, not an outer-test score. Neural fits use source weights 1 and 3, respectively. Candidate screening is followed by repeated inner validation, longer-training challenges up to 1,200 epochs and a separate width challenge. Each family has a locked configuration and training schedule for each outer partition. Final fits restart from their specified initialization, restore the inner-validation rows to training, and average seeds 11, 29, 47, 71 and 101.

Matched no-response fits use the same recipe and schedule. The ExtraTrees and frozen-encoder/tree controls use seeds 11, 29 and 47; common-three-seed comparisons are also reported. Supplementary Methods S18 specifies the search and its limits.

MoLFormer representations are generated with the encoder’s deterministic evaluation option in addition to evaluation mode. An audit found that evaluation mode alone left stochastic attention projections active in the earlier cache. We regenerated all study and stress representations from the same checkpoint and repeated both the neural-head and fixed-tree comparisons. Only these corrected MoLFormer results enter the manuscript’s comparisons. Reload, repeated-query and reordered-batch checks verify prediction replay; the packaged ExtraTrees endpoint does not use MoLFormer and is unaffected. The discrepancy and corrected hashes are recorded in Supplementary Methods S19 and Data 6.

TabPFN-3.5 uses the pinned full and Fast checkpoints with `tabPFN` 9.1.0, run locally under the provider’s non-commercial license. We do not fine-tune the foundation model. Eight candidates combine two checkpoints with four feature recipes; training-side challenges compare eight versus 32 ensemble passes and single versus repeated ARROW context rows. Context repetition represents source emphasis but is not mathematically equivalent to weighted neural loss. Splits precede repetition. The pretrained model’s native preprocessing, including feature selection, is retained; this is not an entirely tree-free preprocessing pipeline. Fitted contexts contain training labels and are not redistributed.

Size, series and predictive-uncertainty diagnostics

The size diagnostic holds out the largest 15% within each source, with stable identity ordering for ties. Every model uses the same 1,160/205 split and three seeds; no held-out label is used in that refit. Configurations come from the locked external-fit recipe. Because the held-out labels were accessible in earlier random-fold development, this is an exploratory conditional test of size transfer. The source models and their source-level out-of-fold predictions are not rebuilt around this size split; exclusion from the endpoint refit does not imply exclusion from all source supervision (Supplementary Methods S5 and S20). The unlabelled grid comprises 30 linear alkanes and 13 members each of capped glycine and alanine series, including the common zero-residue molecule. Series identities, source overlap, fingerprint counts and all predictions are released.

For uncertainty evaluation, we transform each TabPFN distribution to the common raw-target scale and form an equally weighted mixture across five seeds. Central 80% and 95% intervals are quantiles of this mixture, not averages of seedwise quantiles. Empirical coverage is the fraction of external labels inside each interval. We also report widths and verify the mixture mean against stored point predictions. The external cohort is used only to evaluate these intervals, not to calibrate them. Supplementary Methods S20–S21 details these diagnostics.

Statistics

MAE in kcal mol⁻¹ is primary. Root mean square error (RMSE), median absolute error and the fraction of molecules improved are secondary. All comparisons are paired by molecule. Ninety-five percent percentile intervals use 100,000 molecule bootstrap resamples with seed 20260828. For repeated partitions, each molecule’s absolute error is first averaged across repeats before resampling. A predeclared comparison is positive when the complete paired interval for candidate-minus-control MAE is below zero, negative when above zero and neutral otherwise. We report all prespecified comparisons without multiplicity adjustment and do not treat the panel as a single confirmatory hypothesis test. No post-result feature selection is used within the confirmatory comparisons. The later architecture search uses training-side validation and a locked selection policy; its test intervals are descriptive, conditional on fitted models and not adjusted for the many architectures compared.

Deployment artifact

After evaluation was frozen, the response surrogates were fitted with standardized benchmark equivalents removed and the endpoint was refitted on the 1,280 public plus 85 ARROW labels. This all-data artifact is used for deployment and the endpoint-disjoint external evaluation; ARROW accuracy uses only the appropriate fold-specific fits. Inference parses the supplied SMILES with RDKit and converts it to canonical isomeric SMILES before computing descriptors and evaluating the six frozen response-surrogate artifacts; three endpoint regressors are averaged. This step is not a claim of invariance to tautomer, protonation or salt form. The runtime does not read an experimental value, family label, scaffold, fold identifier, trajectory or PIMD result. Software versions, hashes and a SMILES-only smoke test are reported in Supplementary Methods S12–S13 and Tables 7–8.

Acknowledgments

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Competing interests

The authors declare no competing interests.

Use of artificial intelligence

Generative AI tools assisted with code development, manuscript editing and figure preparation. The authors take full responsibility for the content of this work.

Data availability

All benchmark records, held-out predictions, split assignments, audit tables and derived source manifests required to reproduce the paper are provided in the SolvAI repository (Supplementary Data 1–6). The Supplementary Data directory lists the exact filenames and contents of each numbered group. Large third-party datasets are not redistributed where their size or license makes that inappropriate; immutable URLs or DOIs, hashes, licenses and exact preprocessing instructions are supplied in `repro/DATA_PROVENANCE.md`.

Code availability

Source code, the packaged inference artifacts, confirmatory analysis scripts, figure generation, manuscript source and a locked environment are available at <https://github.com/Scientific-Computing-Lab/SolvAI>. A frozen-artifact path reproduces predictions, metrics, figures and PDFs without rerunning physical calculations. The endpoint-comparison directory additionally provides predictions, selection specifications and analysis code. TabPFN weights require the provider’s license; fitted contexts that contain non-redistributable training labels are not published. The packaged public inference command uses the ExtraTrees endpoint, and its runtime measurements do not describe the alternative endpoints.

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Supplementary Information

Supplementary Methods S1–S21; Supplementary Notes S1–S2; Supplementary Tables 1–16; Supplementary Figures 1–12; Supplementary Data 1–6.

Supplementary Methods

S1. Reference-set processing

The ARROW reference table was reconstructed from machine-readable project outputs and contains 85 unique neutral solutes in water at 298 K. One molecule–condition row was retained for each connectivity. The reference includes experimental hydration free energy and, where available in the source, uncertainty, classical ARROW, PIMD4 and PIMD8 values. Molecular names were not used for identity. Canonical isomeric SMILES, full InChIKey and the first InChIKey block were generated with RDKit 2026.03.5. The released benchmark table is `data/benchmark/arrow_solvation_master.parquet`.

The reconstructed ARROW/PIMD8 MAE is 0.204836 kcal mol^{−1} and the classical ARROW MAE is 0.784648 kcal mol^{−1}. These values are direct errors against the same experimental column used for SolvAI evaluation; neither simulation value is an input to the final model.

S2. Public endpoint labels

The endpoint-label catalog initially contained 5,075 benchmark-disjoint connectivities. A frozen source-quality rule retained all 1,147 rows from an earlier FreeSolv/CombiSolv-Exp merge and 133 additional SoluteML dGsolvDB3 rows supported by at least two source measurements, giving 1,280 public labels. Duplicate structures were resolved at the connectivity level before this rule was applied. The source composition is listed in Supplementary Table 3. The raw merged tables are not redistributed where source terms are not explicit; URLs, versions, hashes and reconstruction commands are given in `repro/DATA_PROVENANCE.md`. CombiSolv-Exp originates from the collection of Vermeire and Green;⁹ the version used here was downloaded from the SolvBERT repository mirror.⁴¹

For a random outer fold, these 1,280 rows were combined with the available ARROW outer-training molecules. Public rows received sample weight 1 and ARROW outer-training rows weight 3. The experimental value of an outer-test molecule was never used. In the zero-ARROW transfer experiment, only the 1,280 public labels were fitted and all 85 ARROW molecules were evaluated.

The weight of 3 was part of the prospective confirmatory protocol rather than a post-result choice. A separately prespecified one-factor sensitivity changed only the available ARROW outer-training weight to 1. Under this setting, matched structure-only and response-augmented ExtraTrees MAEs are 0.30977 and 0.20642 kcal mol^{−1}, respectively; the paired change is −0.10335 kcal mol^{−1} (95% interval, −0.21883 to −0.01995).

S3. Response-source processing

CombiSolv-QM. The water subset was deduplicated by solute connectivity and expressed in kcal mol^{−1}. After the original exact-connectivity exclusion, 3,961 structures were available. Conservative canonical-tautomer comparison found two additional benchmark equivalents; the confirmatory surrogate therefore used 3,959 rows.

SoluteML Abraham parameters. Five Abraham solute axes (E, S, A, B and L) were predicted from 8,098 benchmark-disjoint structures. These coordinates respectively describe excess molar refraction, dipolarity/polarizability, hydrogen-bond acidity, hydrogen-bond basicity and gas-to-hexadecane partition response. No standardized benchmark equivalent was found in the audited source.

OpenFF explicit-water response. The source contains 520 benchmark-disjoint absolute solvation-free-energy records from an OpenFF 2.3.0/NAGL/TIP3P protocol with 14 alchemical states. The records come from the Open Free Energy (OpenFE) alchemical benchmark resources release,⁴² using version 2.3.0 of the OpenFF Sage force-field family.²⁶ Separate ExtraTrees models predicted calculated hydration free energy and its experimental residual, defined as experimental hydration free energy minus the original calculated hydration free energy, not minus a surrogate prediction. Their sum is the retained corrected OpenFF coordinate.

GBn2 implicit response. The source contains 550 benchmark-disjoint GBn2 hydration calculations. GBn2 is the generalized Born model of Nguyen, Roe and Simmerling;²⁷ the calculation records were obtained from the dataset released by Katzberger and Riniker.⁴³ Separate ExtraTrees models predicted the calculated value and its experimental residual. The sum is the retained corrected implicit-solvent coordinate. Its residual target uses the same experiment-minus-original-calculation definition.

MolSolv SMD(water). The source archive contains 1,729,545 M06-2X/6-31G* SMD calculations. The frozen selection retained all eligible neutral structures with at most 10 heavy atoms and a deterministic one-in-eight sample for structures with 11–18 heavy atoms. After deduplication and exact-connectivity filtering, 350,391 rows remained. Canonical tautomer standardization identified 32 additional benchmark-equivalent records. The confirmatory model used 350,359 rows while keeping every retained molecule in its original source split.

ConfSolv H₂O. The archive contains 5,392,567 H₂O conformer records for 43,693 solutes. We converted kJ mol⁻¹ to kcal mol⁻¹ and aggregated conformers into gas, water and hydration conformer corrections together with water-response distribution summaries. The broader quality filter produced 39,878 neutral connectivities. The selected surrogate target set had 17,851 rows with complete features and admissible energy range; fragment-parent, uncharging and tautomer standardization removed 22 additional benchmark-equivalent records, leaving 17,829.

S4. The 15 response descriptors

Supplementary Table 1 defines every retained coordinate. At inference, the table’s “surrogate” column is the calculation that runs; the expensive source method does not. The final vector contains one CombiSolv-QM response, five Abraham axes, one corrected OpenFF response, one corrected GBn2 response, one SMD(water) response and six ConfSolv summaries, for 15 coordinates in total.

Richer CombiSolv, OpenFF, GBn2 and ConfSolv targets were available and were evaluated during model development. Only the listed coordinates belong to the released stack. In particular, no PIMD2, PIMD4, PIMD8, dH/dλ or NQE coordinate is present.

S5. Surrogate models

The CombiSolv-QM and MolSolv targets were fitted with Chemprop 2.2.4 directed-message-passing models initialized from CheMeleon. Both used a two-layer feed-forward head of width 256, dropout 0.1, initial/maximum/final learning rates of 3×10^{-5} , 10^{-4} and 10^{-5} , and PyTorch seed 20260826. CombiSolv-QM used RDKit two-dimensional descriptors, batch size 64, a 30-epoch cap and patience 7; MolSolv used the molecular graph, batch size 256, a 25-epoch cap and patience 6. The original frozen train/validation/test memberships were 80/10/10% for CombiSolv-QM and 90/5/5% for MolSolv. Standardized equivalents were removed from their existing split rather than resplitting the remaining source.

Abraham, OpenFF and GBn2 targets used ExtraTrees regressors on the same 2,265 structure features used by the endpoint. The selected leaf sizes and target-level OOF errors are recorded in the source metadata. ConfSolv used 217 RDKit descriptors and the 256 highest-variance Morgan bits. Each target was fitted with a LightGBM L1-regression model with 450 estimators, learning rate 0.04, 31 leaves, minimum child size 20, row subsampling 0.8, column subsampling 0.55 and L_2 regularization 0.1. Target-specific seeds were 20260826 plus the target index for validation and 1000 plus that index for the all-retained-source fit.

For OpenFF and GBn2, source records were matched to public endpoint candidates by connectivity. Public molecules represented in a source table received predictions from five-fold source-level out-of-fold fits for each physical and residual target; other public molecules and ARROW molecules received the corresponding full-source model predictions. Minimum leaf size was selected from 1, 2 and 3 using those same source folds. Thus the recorded source OOF errors are selection scores, not nested unbiased validation estimates. These source folds were fixed independently of the later endpoint partitions; they were not rebuilt for chemical- or size-held-out endpoint fits. Source-OOF feature generation therefore differs from excluding an entire downstream holdout from source supervision. The source-preparation scripts are archived under `repro/response_sources/`.

The recorded OpenFF physical/residual OOF MAEs are 0.617/0.531 kcal mol⁻¹; the corresponding GBn2 values are 0.669/0.626. These are errors of separate source components, not of their corrected sum. The standardized-exclusion ConfSolv validation contains 1,785 molecules and gives MAEs of 0.248, 0.267, 0.354, 0.448, 0.688 and 0.450 kcal mol⁻¹ for coordinates 10–15 in Supplementary Table 1. These heterogeneous source checks do not form a common independent fidelity benchmark for all 15 coordinates; the endpoint comparisons assess their predictive utility rather than presuming uniform recovery accuracy.

S6. Endpoint model

The molecular representation comprises a 2,048-bit radius-2 Morgan fingerprint and 217 RDKit two-dimensional descriptors. The confirmatory endpoint appends the 15 response descriptors, giving 2,280 columns. A median imputer is fitted on each training partition and adds missingness indicators only where required. Each of three ExtraTreesRegressor models uses 360 trees, squared-error splitting, `max_features=0.7`, `min_samples_leaf=2`, `min_samples_split=2`, unlimited depth and no bootstrap; seeds are 11, 29 and 47. Predictions are averaged. The matched structural model removes only the 15 response columns.

After all evaluation was frozen, the release artifact was refitted on the 1,280 public labels and all 85 ARROW labels using the same weights and configuration. That refit supplies deployment predictions and the endpoint-disjoint external evaluation. Reported ARROW held-out scores use only the appropriate fold-specific fits.

S7. Validation partitions

The fixed primary result uses the released `fold_random` assignment. Five complete repeated five-fold partitions use shuffled K-fold seeds 314159, 271828, 161803, 141421 and 173205. All comparisons use the same fold, training rows and model seeds within a partition.

For functional-family separation, SMARTS rules were applied in the precedence order carboxylic acid, amide, ester, aldehyde, ketone, alcohol/phenol, ether, amine, thiol, sulfide/disulfide, aromatic heterocycle, carbocyclic aromatic, alkene/diene, saturated hydrocarbon and other. A test-fold family was removed from both public and ARROW endpoint training rows. Bemis–Murcko scaffold separation used canonical isomeric scaffold SMILES; acyclic molecules were keyed by primary family. Molecular clusters were constructed once on the combined endpoint pool with Taylor–Butina clustering at Morgan Tanimoto similarity 0.70. Nearest-neighbor analyses excluded training rows at similarity thresholds 0.50, 0.60, 0.70 and 0.80 to any outer-test molecule.

S8. Shuffled-descriptor control

All 15 response columns were permuted together to preserve their joint distribution. For each outer fold, public, ARROW outer-training and ARROW outer-test response rows were permuted separately. The experimental target was never used to define a permutation. Five base seeds (88001–88005) were used for the fixed partition and each repeated partition. This control changes molecule–response alignment while retaining feature count, response marginal distributions and inter-descriptor correlations.

S9. Statistical analysis

MAE is primary. RMSE, median absolute error and the fraction of molecules improved are secondary. Paired candidate-minus-reference differences were resampled at molecule level 100,000 times with seed 20260828. The 2.5th and 97.5th percentiles define the reported intervals. For the five repeated partitions, each molecule’s absolute error was first averaged over partitions before paired resampling. The prespecified protocol defined a positive comparison as a complete interval below zero. We report all prespecified comparisons without multiplicity adjustment and do not treat the panel as a single confirmatory hypothesis test. The protocol and its implementation amendments are archived in Git at `release/CONFIRMATORY_FREEZE.md`; the amendments preceded the corrected runs. For the additional molecular-model comparisons, the released `results/endpoint_models_20261006/corrected_original_metrics.csv` also provides the coefficient of determination (R^2) for each model and cohort.

S10. External validation

The original 221-molecule cohort was frozen for a separate solvation-model validation study before that study’s model predictions. It was derived from type-M original-measurement records in Sander’s Henry-law compilation v5.0.0. Intrinsic pure-water H_s^{cp} values (aqueous concentration divided by gas partial pressure), normalized to 298.15 K, were converted as $K_c = H_s^{cp} RT$ and $\Delta G_{hyd} = -RT \ln K_c$, where K_c is the dimensionless aqueous-to-gas concentration ratio. This produces the same 1 M gas-to-1 M aqueous standard-state convention used for the endpoint. The source values are printed to two significant figures.

SolvAI qualification was performed without reading SolvAI predictions or errors. All 221 rows passed the connected, neutral, supported-element and target-compatibility rules. One candidate was excluded because that study’s registry audit had documented a definite conflict between its registry name/CAS and supplied InChIKey structure. The remaining 220 molecules had no exact, fragment-parent, uncharged-parent or canonical-tautomer equivalent among the 1,280 public or 85 ARROW endpoint labels. Of these, 123 occurred in at least one supervised response-source table and 97 were absent from all six. The latter define the strict response-source-disjoint subset. The cohort hashes, all exclusions and all response-source matches were recorded in Git before endpoint evaluation.

Both endpoint models were then fitted deployment-style on the same 1,280 public plus 85 ARROW labels with weights 1 and 3. They used the same descriptors, imputation, three 360-tree ensembles and seeds; only the 15 response columns differed. No external-cohort label entered fitting or selection. On $N=220$, structure-only and response-augmented ExtraTrees MAEs were 1.53165 and 1.15255 kcal mol^{−1} (paired change -0.37909 ; 95% interval, -0.51692 to -0.24817). On strict $N=97$, the corresponding values were 2.13830 and 1.53560 kcal mol^{−1} (paired change -0.60271 ; interval, -0.86277 to -0.35858). The absolute errors delimit applicability and are not presented as PIMD8-level performance outside ARROW-85.

S11. Lambda-response probes

Short PIMD2 response fingerprints were available at $\lambda = 0.1, 0.5$ and 0.9 for 72, 74 and 73 molecules, respectively; 72 had complete three-point curves. The targets were the total solute-solvent $dH/d\lambda$ and its electrostatic and van der Waals components. A zero-valued polarization field was not interpreted. Structure-only heads were evaluated by OOF prediction, and their outputs were added to the same matched endpoint used for the corresponding exploratory ablation. A separate direct-integration experiment used the predicted three-point curve with an affine calibration fitted inside each outer training fold. These targets were evaluated as candidate supervision and were not retained.

S12. Artifact verification

The public API accepts one or more SMILES strings. RDKit parses each input and emits canonical isomeric SMILES before feature generation. It does not standardize a query across tautomers, protonation states or salt forms. The runtime computes descriptors, evaluates the two D-MPNNs and four tree-artifact families, and passes the resulting 2,280-vector to three endpoint regressors. The artifact audit checks file hashes, feature count and names, software versions and two smoke predictions. It rejects features containing experimental target, PIMD, trajectory, probe, fold or scaffold information. The standardized-exclusion surrogate refit is identified in the model card and manifest. The ensemble spread is returned for diagnosis but is not a calibrated per-query applicability or reliability score.

S13. Runtime benchmarking

Runtime was measured end to end from input SMILES to prediction on the released artifact. The host used an Intel Core i7-4930K CPU; the installed NVIDIA RTX 3090 was not used for inference. Cold single-molecule latency, five warm single-molecule runs, three batch-32 runs and peak resident memory were recorded. Chemprop predictor startup dominates single-molecule latency. These measurements describe the released implementation, not the irreducible model cost. The ARROW/PIMD8 comparison uses the published 15-window protocol and is kept separate from same-host inference timing.

S14. Additional molecular-model comparisons

The additional molecular-model comparison uses the existing 1,280 public endpoint labels, 85 ARROW labels and 220 external test molecules. The protocol was fixed before the new fits, but these are retrospective comparisons on already examined datasets. The original fixed five-fold ARROW assignments are retained. For the external test, all 1,365 endpoint labels are used; the 97-molecule source-disjoint subset is fixed before these comparisons and is not a separate fit. Public labels have weight 1 and ARROW training labels weight 3 throughout. The wider optimization study in S18 is a separate comparison, not a reinterpretation of these fixed 80-epoch fits.

The scratch D-MPNN has message dimension 300, three message-passing steps, mean aggregation, a regression head with two hidden layers of 300 units and dropout 0.1. The CheMeleon model initializes message passing from the published `chemeleon_mp.pt` weights, retains their encoder configuration and fine-tunes that encoder; it uses the same regression-head dimensions and dropout. Both optimize squared error using Adam and batch size 64, with constant learning rates 3×10^{-4} and 10^{-4} , respectively. Target scaling is fitted on the training subset only. For each outer partition and model seed, a deterministic 15% of each source is held out internally (split seed 20261002 + model seed). The epoch with lowest unweighted validation MAE is selected, with a cap of 80 epochs and patience 12. A model reinitialized from the same random or CheMeleon starting weights is then fitted on all outer-training labels for that many epochs, using a newly fitted training scaler. This restores the inner-validation molecules to the final training pool without exposing the outer test labels. Predictions from seeds 11, 29 and 47 are averaged. Every selected epoch, validation trace, training identity and held-out identity is retained. Supplementary Fig. 6 shows the order of inner selection, fresh refitting and held-out scoring.

MoLFormer uses the published `ibm-research/MoLFormer-XL-both-10pct` model at revision `7b12d946c181a37f6012b9dc3b002275de070314`. The 768-dimensional representation is the attention-mask mean of the last hidden state, with maximum sequence length 256. Uni-Mol uses its 512-dimensional molecular representation from the `mol_pre_all_h_220816` checkpoint, including hydrogens. Its RDKit conformer generator uses seed 42. MoLFormer uses the corrected deterministic representations described in S19; Uni-Mol reuses its original label-free cache, with missing structures encoded by the same pipeline. Three of the 353 structures absent from the original Uni-Mol embedding cache used the package’s two-dimensional fallback after three-dimensional embedding failed; they are retained rather than excluded by prediction quality. Both pretrained representations are frozen during endpoint fitting. Molecular membership in their original pretraining datasets is not audited here.

For each pretrained representation, we fit the same three 360-tree ExtraTrees regressors on structural features plus that representation, then repeat with the 15 SolvAI response descriptors added. Imputation is

fitted inside each training partition. All families, both feature conditions and all seeds are retained in the report. The comparison is limited to these fixed training configurations; it is not an exhaustive hyperparameter optimization or a claim to the best achievable performance of an architecture. Code and hashed inputs are provided with `scripts/run_journal_baselines.py`; independent aggregation is provided with `scripts/summarize_journal_baselines.py`.

S15. Conformational meaning of the response targets

The experimental endpoint is the equilibrium hydration free energy of a molecular species. It does not assign a separate target to each geometry of that species. The ConfSolv H₂O subset supplies conformer-resolved gas, water and solvation free energies, which are aggregated before surrogate training. For gas or water conformer energies G_i , define $g_i = G_i - \min_j G_j$. The conformer correction is $C = -RT \ln \sum_i \exp(-g_i/RT)$. Two response descriptors are the gas and water corrections. The third is the difference between the ensemble hydration estimate and the solvation energy of the lowest-gas-energy conformer. The remaining three are the population standard deviation of conformer solvation energies and the arithmetic mean and population standard deviation of $g_i^{\text{water}} - g_i^{\text{gas}}$. Source energies are converted from kJ mol⁻¹ to kcal mol⁻¹. These six summaries characterize the supplied conformer set; they do not establish complete conformational sampling or provide a geometry-specific prediction at inference. Supplementary Fig. 7 illustrates these six summaries and the distinction between individual conformers and the molecular equilibrium endpoint.

S16. Error as a function of chemical similarity

For each external molecule, we use the maximum Tanimoto similarity of its radius-2, 2,048-bit Morgan fingerprint to any of the 1,365 endpoint-training molecules. Fixed descriptive bins are [0, 0.3), [0.3, 0.5), [0.5, 0.7) and [0.7, 1]. We report counts, MAEs and paired differences within each occupied bin. These are descriptive strata of an existing cohort, not an independently calibrated applicability-domain threshold. In particular, similarity to endpoint training and membership in a response-source dataset are different quantities. The exact molecule-level similarities, predictions and bootstrap intervals are supplied in `results/journal_20261002/`.

S17. Computation-only response control

A separate source-composition analysis combines ten descriptors derived solely from computed source targets: COSMOtherm water free energy, uncorrected OpenFF and GBn2 free energies, SMD(water), and the six ConfSolv summaries. It excludes all five empirical Abraham parameters and both experimental residual corrections. The ten descriptors are added to the same 2,265 structural features, with unchanged five-fold ARROW evaluation, endpoint labels, sample weights, trees and three seeds. The comparison was specified before fitting this additional control. It measures whether computed-source descriptors alone improve this learner; it is distinct from the original three-descriptor computation-core ablation. After the ARROW comparison, we specified an extension of the same fixed ten-descriptor model to the external cohort. The OpenFF and GBn2 components are recovered directly from the original packaged surrogates. Before fitting, their corrected sums must reproduce the frozen external response cache to within 10⁻¹⁰ kcal mol⁻¹. All other response descriptors come from that cache. The external fit uses all 1,365 endpoint labels with the original weights and seeds. Both stages, exact predictions and input hashes are retained; the extension is explicitly retrospective.

S18. Endpoint architecture and optimization study

The study reuses the 1,280 public labels, 85 ARROW labels, 220 external identities and original five-fold ARROW assignments. Source-surrogate weights, targets and experimental labels are unchanged. It is retrospective, with adaptive development before a final selection lock; it is not a new prospective validation. Final outer predictions are made only after training-side selection is complete. The common selection score averages public-source and ARROW-source validation MAE equally. Three random inner splits each hold out 15% of each eligible source, using seeds 640601 + j for $j = 0, 1, 2$. Public training rows have loss weight 1 and ARROW rows weight 3. No external label participates in these fits or inner decisions.

The initial 52-candidate list covers residual MLPs, dual-branch networks, ridge-plus-neural residuals, pretrained-encoding heads and directed message passing. Tabular residual blocks apply layer normalization, two linear transforms with GELU activation and dropout, then add an identity skip. The dual architecture keeps a separate 32-unit response branch before concatenating it with the structural branch. The ridge model fits a training-only linear baseline with regularization 10 to the scaled RDKit and response columns, then trains a residual network on the remaining target. This is not a ridge model on seven selected responses with a tree residual. The graph variants use Chemprop bond message passing with either summed and mean atom embeddings plus atom count, or summed learned atomic outputs plus a global correction. These readouts carry size information

but do not impose exact additivity of the full free-energy predictor. Frozen-encoder neural heads use separate structure/encoding and response branches.

Candidate configurations vary widths 32–256, depth, dropout 0–0.2, learning rates from 3×10^{-4} to 3×10^{-3} , weight decay, squared or Huber loss, and plateau-reduced or cosine schedules. AdamW and batch size 128 are used. Ordinary neural fits have a 600-epoch cap. Median imputation, constant-column removal and numerical scaling are fitted inside each training split; response columns are retained even if constant. Binary Morgan bits remain unscaled. The skewed Ipc descriptor is transformed by $\log(1 + \max(\text{Ipc}, 0))$ or removed; BertzCT and Kappa1–3 use signed-log transforms. Physical counts, mass and surface descriptors are not clipped to the training range. Count-based fingerprints and later raw-count, log-count and no-fingerprint variants test whether loss of multiplicity in binary Morgan features limits size response.

Each candidate is screened on split 0 at seed 11. The best two per family proceed to three splits and seeds 11/29. The family winner is challenged with a 1,200-epoch cosine schedule on the same three splits at seed 47, against its ordinary schedule. The longer schedule is retained only if mean validation MAE decreases by more than 0.005 kcal mol⁻¹ and at least two of three pairs improve. A retained long schedule is completed at seeds 11/29, giving nine inner fits. Thus this challenge changes both duration and learning-rate trajectory, not duration alone. Fresh outer fits use the median selected epoch and the inner-derived schedule. A width boundary challenge expands selected 256-unit models to 512 units when a family selected the upper width in at least four of six partitions; graph message dimension is expanded as well. The same paired improvement rule determines retention.

TabM adds 12 configurations spanning three fingerprint treatments and four width, depth, dropout and numerical-embedding settings. Each model uses 32 parameter-sharing ensemble members; their mean is the prediction.³² The study uses TabM 0.0.3 and rtdl-num-embeddings 0.0.12. Representation refinements and the width extension are retained as explicit selection variants, not relabelled as separate architectures. Nine of 24 width challenges passed to nine-repeat completion. All 81 retained wider-fit optima occur before 95% of their training horizon. Longer schedules were selected in all six corrected-MoLFormer partitions, five dual and TabM partitions, three graph partitions, and two residual-MLP and ridge partitions. Supplementary Fig. 12 shows the paired duration and width diagnostics.

The search contains 1,812 inner evaluation records, including 174 TabPFN context evaluations. Gradient-based runs total 1,018,212 epochs; TabPFN’s 2,112 internal ensemble passes are not gradient epochs. Summed fitting time is 45.06 hours across overlapping workers, not elapsed time or GPU-hours. The defined search and boundary checks completed, but do not exhaust all architectures, regularization choices or training-data interventions. Full candidate lists and per-partition decisions are in Supplementary Data 6. Final comparisons retain every family, not only the global selector. Matched no-response models inherit the response model’s recipe, so the comparison does not estimate each structural-only architecture’s best achievable error.

S19. Pretrained representations and TabPFN configuration

The earlier MoLFormer cache used evaluation mode without enabling the model’s `deterministic_eval` option. Stochastic attention projections remained active. All 1,585 study representations and 56 stress-series representations were therefore regenerated from the same checkpoint with both settings enabled, and all valid MoLFormer endpoint comparisons were refitted. Identical repeated encoding gave zero change; changes in query order and batching produced representation differences below 3×10^{-5} . Fresh encoding through endpoint prediction agreed within 4.9×10^{-6} kcal mol⁻¹. Corrected fixed-tree results replace the earlier cache-based values in Fig. 5 and Supplementary Tables 11–12. Legacy stochastic-cache branches remain labelled in the released diagnostic data and are excluded from valid-model selection and the manuscript’s performance comparisons. Uni-Mol’s conformer pipeline and the packaged ExtraTrees endpoint are unaffected. Original encoder pretraining membership remains unaudited.

TabPFN-3.5 uses `tabpfn` 9.1.0 and full/Fast checkpoints from revision `06bf2ba35c80a92a3b9abb436b99cf49e7a0365e` of `Prior-Labs/tabpfn_3_5`.³⁴ Checkpoint SHA-256 values are recorded in Supplementary Data 6. This specific software release is distinguished from the earlier tabular foundation-model paper.³³ We use its pretrained weights without fine-tuning. Inputs retain training-fitted native preprocessing, including feature subsampling/ranking and target transformations. The use of a transformer endpoint therefore does not mean that all preprocessing or the six source-surrogate families are neural.

Eight candidates combine full/Fast weights with four recipes: RDKit plus binary Morgan bits; RDKit plus raw Morgan counts; RDKit alone; or RDKit plus the corrected MoLFormer representation. Each has the 15 responses in its full condition. The best two screen candidates receive repeated inner evaluation. Training-side challenges compare one versus three copies of each eligible ARROW context row, and eight versus 32 ensemble passes, using the same greater-than-0.005 and two-of-three improvement rule. Context repetition occurs after splitting and does not create new independent observations or equal weighted gradient fitting. All selected recipes use the full checkpoint and three ARROW copies. The selected ARROW folds 0–4 use respectively raw counts/8 passes, no fingerprint/32, no fingerprint/8, raw counts/32, and MoLFormer without fingerprint/8. The

external fit uses MoLFormer without fingerprint and eight passes. These are a per-partition selection policy, not one identical input recipe across folds.

The comprehensive policy selects the eligible option with lowest repeated inner score independently in each partition. It selects TabPFN in all six partitions. Five-seed final predictions use seeds 11, 29, 47, 71 and 101; controls remove only the responses. Common-three-seed means address the unequal ensemble size relative to the three-seed ExtraTrees fits. Every fitted-state replay and final identity/label audit passed. Licensed TabPFN weights and fitted contexts are retained privately; contexts embed training labels and are not public model-weight substitutes.

S20. Size holdout, homologue series and label influence

The size split ranks molecules by descending heavy-atom count within public and ARROW sources, breaking ties by identity. The largest 15% comprise 192 public and 13 ARROW molecules, leaving 1,160 training rows. Training heavy-atom counts range from 0 to 17 and holdout counts from 8 to 37; 188 holdout molecules exceed the global training maximum. Models use their locked external-fit recipes and three fresh seeds. ExtraTrees is refitted on these 1,160 rows rather than evaluated using release weights that have seen the held-out labels. No size score affects selection. Since the held-out labels participated in earlier random-fold development, intervals describe this exploratory split, not prospective selection-free performance on new molecules. Supplementary Table 15 gives all valid family results. The 13 ARROW molecules retain their original exclusion from supervised source tables. No analogous source-disjoint guarantee is made for the 192 public molecules: source models and source-OOF features remain fixed, with no size-specific source refitting or exposure audit. Accordingly, this is a conditional endpoint comparison, not a validation of fully source-unseen size extrapolation.

The unlabelled grid contains linear C1–C30 alkanes and Ac-(Gly) $_n$ -NHMe and Ac-(Ala) $_n$ -NHMe for $n = 0, \dots, 12$. The two zero-residue rows represent the same N-methylacetamide molecule, so 56 rows are not 56 unique structures. All positive-length peptides are absent from endpoint training; N-methylacetamide is present. The longest glycine and alanine examples contain 53 and 65 heavy atoms, respectively, beyond the maximum 37 in the original training pool. Binary Morgan fingerprints are identical across $n = 2, \dots, 12$ within either peptide family and across C7–C30 alkanes. Count fingerprints, RDKit descriptors and some predicted responses still change, so the entire input does not freeze. An initial neural failure involved very large Ipc excursions; the present transformed-input study avoids that numerical failure without establishing physical accuracy.

For ExtraTrees, each leaf value is an average of eligible training labels and the ensemble output is an average of leaf values. Averaging each tree’s minimum and maximum leaf values gives an outer prediction bound, not necessarily a jointly attainable extremum. The released ensemble bound is $[-32.5945, 5.7565]$ kcal mol $^{-1}$. TabPFN’s mean is a convex combination of finitely many bin means on its fitted raw-target scale; averaging seedwise extrema gives the wider outer bound $[-658.4946, 649.9221]$. Its edge distributions can have unbounded tails without an unbounded mean. Changing this bound is not a proof of size extensivity.

Only C17, C29 and C30 are absent from endpoint training. C14 and C16 labels of 2.16 and 1.22 kcal mol $^{-1}$ are faithful rounded means of SoluteML’s linked CompSol records (C14: 68956/68957; C16: 67229/67230). The revised CompSol archive retains these binary-mixture records. Critical IUPAC–NIST evaluations identify disagreement in the underlying solubility measurements and do not recommend a best solubility value (https://srdata.nist.gov/solubility/sol_detail.aspx?sysID=38_354 and https://srdata.nist.gov/solubility/sol_detail.aspx?sysID=38_389). We did not convert an extrapolated trend into a replacement label. A same-recipe three-seed refit omitting only C14/C16 raises their predictions from 2.858/2.206 to 4.118/4.204 kcal mol $^{-1}$. This isolates substantial label influence but does not establish that the refitted predictions are correct. All principal architecture comparisons retain the original labels.

S21. Native predictive intervals

For the five final external TabPFN fits, each predictive distribution is expressed on the same raw kcal mol $^{-1}$ grid. Equal-weight mixture cumulative probabilities are inverted to obtain central 80% and 95% intervals, rather than averaging individual quantiles. Reconstructed means agree with stored predictions within 1.4×10^{-5} kcal mol $^{-1}$. The external 220 labels and strict 97 subset are used to assess, not calibrate, coverage. At 80%, 128/220 and 54/97 labels are covered; at 95%, 176/220 and 74/97 are covered. Supplementary Table 16 also reports interval widths. These undercoverage rates preclude describing the native intervals as calibrated reliability estimates. Seed dispersion and bootstrap intervals for mean errors are different quantities. No peptide coverage can be measured without reference free energies.

Supplementary Notes

S1. Identity relationship to FreeSolv

Eighty of the 85 ARROW reference connectivities have a FreeSolv identity. This fact does not establish the provenance of every experimental value in the ARROW study. It instead determines the leakage rule: those connectivities were removed from all external supervised sources before endpoint or surrogate fitting. The frozen SolvAI artifact is therefore not evaluated over unfiltered FreeSolv and no generic FreeSolv state-of-the-art claim is made.

S2. Interpretation of alternative supervision

Non-improving features are not interpreted as evidence that their underlying physical methods are inaccurate. A response can fail downstream because it is poorly predicted from structure, redundant with existing coordinates, mismatched to the endpoint or supported by too few labels. Compact ConfSolv summaries outperform larger graph and feed-forward latent representations in the matched exploratory screen. Sparse PIMD2 response produces physically recognizable trends but not the precision required for hydration-free-energy integration. These observations motivate the three considerations stated in the main text: relevance, structural learnability and non-redundancy. The screens do not isolate their individual causal effects.

Supplementary Data

The first five numbered groups are supplied under `paper/supplementary_data/` in the repository and source-data archive. CSV files are the archival tables; XLSX files provide corresponding worksheets. The filenames below are relative to that directory.

Supplementary Data 1: experiment ledger.

`Supplementary_Data_1_experiment_ledger.csv` and the same stem with `.xlsx` record the evaluated scientific experiments, including non-retained branches.

Supplementary Data 2: molecule-level predictions.

`Supplementary_Data_2_molecule_predictions.csv` and `.xlsx` contain experimental values, ARROW/PIMD8 values and held-out SolvAI predictions.

Supplementary Data 3: validation assignments.

`Supplementary_Data_3_split_assignments.csv` contains molecular identities, family, scaffold and cluster assignments; the `.xlsx` file additionally contains repeated-partition folds. Primary folds are included in Supplementary Data 2. Complete chemical-exclusion details are supplied in `results/confirmatory/` alongside the analysis outputs.

Supplementary Data 4: response and endpoint provenance.

`Supplementary_Data_4_teacher_priors.csv` defines the 15 descriptors; `Supplementary_Data_4_teacher_sources.csv` and `Supplementary_Data_4_endpoint_sources.csv` describe their sources. `Supplementary_Data_4_teacher_manifests.xlsx` collects these tables and the identity audit. Source versions, licenses and hashes are documented in `repro/DATA_PROVENANCE.md` and the linked manifests. The artifact filenames retain their original identifiers.

Supplementary Data 5: external-cohort validation.

`Supplementary_Data_5_tier_a_predictions.csv`, `Supplementary_Data_5_tier_a_qualification.csv` and `Supplementary_Data_5_tier_a_external_validation.xlsx` provide matched predictions, eligibility decisions and response-source exposure.

Supplementary Data 6: endpoint architectures and size diagnostics.

`results/endpoint_models_20261006/` contains the model/ablation metrics, held-out and unlabelled-series predictions, paired comparisons, size aggregates, native predictive intervals, selection specification and publication manifest. `repro/endpoint_models/README.md` describes reproduction and restricted training-context dependencies. Legacy stochastic-encoding rows are explicitly marked and are excluded from valid comparisons. Corrected original-comparison tables replace the MoLFormer rows without overwriting the earlier source archive.

Supplementary Tables

Supplementary Table 1 | SolvAI response descriptors

Index	Artifact name	Physical meaning
1	combisolv_qm	COSMOtherm water response
2	abraham_e	excess molar refraction E
3	abraham_s	dipolarity/polarizability S
4	abraham_a	hydrogen-bond acidity A
5	abraham_b	hydrogen-bond basicity B
6	abraham_l	gas-to-hexadecane partition L
7	openff_corrected	explicit-water alchemical response
8	gbn2_corrected	implicit-solvent hydration response
9	smd_water	SMD water response
10	conf_gas_corr	gas conformer correction
11	conf_solution_corr	water conformer correction
12	conf_hydration_corr	hydration conformer correction
13	conf_gsolv_sd	conformer solvation-energy spread
14	conf_response_mean	mean conformer solvent response
15	conf_response_sd	conformer response spread

Indices	Source	Surrogate	Training rows	Transformation
1	CombiSolv-QM	CheMeleon D-MPNN	3959	direct
2–6	SoluteML	ExtraTrees	8098	direct
7	OpenFF 2.3.0 ASFE	ExtraTrees	520	prediction + residual
8	GBn2	ExtraTrees	550	prediction + residual
9	MolSolv	CheMeleon D-MPNN	350359	direct
10–15	ConfSolv H2O	LightGBM	17829	direct

Units are kcal mol^{−1} except for indices 2–6, which use their respective Abraham scales. Every descriptor is inferred from structure by its surrogate; no source simulation is run at inference. ASFE denotes absolute solvation free energy. The six conformer summaries are defined explicitly in Supplementary Methods S15.

Supplementary Table 2 | Source-specific confirmatory filtering

source	unit	before standardized exclusion	removed	retained	role
CombiSolv-QM	unique structures	3961	2	3959	COSMOtherm water
MolSolv	unique structures	350391	32	350359	SMD(water)
ConfSolv	model-usable connectivities	17851	22	17829	conformer response
Endpoint labels	experimental connectivities	1280	0	1280	hydration free energy

Supplementary Table 3 | Public endpoint-label sources

source group	selected records	selection rule
FreeSolv only	69	benchmark-disjoint connectivity
CombiSolv-Exp only	588	water records; benchmark-disjoint connectivity
FreeSolv + CombiSolv-Exp	490	one duplicate-resolved connectivity
SoluteML dGsolvDB3	133	at least two source measurements

Supplementary Table 4 | Complete repeated-partition values

MAE is in kcal mol^{−1}; each repeat is a complete five-fold partition.

Repeat	Split seed	Method	MAE
0	314159	Structure only	0.31780
0	314159	SolvAI	0.20788
1	271828	Structure only	0.31059
1	271828	SolvAI	0.21010
2	161803	Structure only	0.31309
2	161803	SolvAI	0.21217
3	141421	Structure only	0.31602
3	141421	SolvAI	0.20618
4	173205	Structure only	0.30727
4	173205	SolvAI	0.20053

Supplementary Table 5 | Global chemical-separation results

MAE is in kcal mol⁻¹. NN denotes the nearest-neighbor similarity exclusion threshold: endpoint-training rows at or above that threshold are removed.

Separation	Method	N	MAE
Functional family	Structure only	85	1.262
Functional family	SolvAI	85	0.468
Scaffold	Structure only	85	0.728
Scaffold	SolvAI	85	0.376
Cluster (similarity 0.70)	Structure only	85	0.349
Cluster (similarity 0.70)	SolvAI	85	0.216
NN exclusion 0.50	Structure only	85	0.359
NN exclusion 0.50	SolvAI	85	0.219
NN exclusion 0.60	Structure only	85	0.344
NN exclusion 0.60	SolvAI	85	0.211
NN exclusion 0.70	Structure only	85	0.343
NN exclusion 0.70	SolvAI	85	0.211
NN exclusion 0.80	Structure only	85	0.339
NN exclusion 0.80	SolvAI	85	0.210

Supplementary Table 6 | Diagnostic family errors

Fixed-partition out-of-fold response-augmented ExtraTrees errors, grouped by diagnostic family; these are not family-held-out predictions. MAE is in kcal mol⁻¹.

Family	N	MAE
Acids	3	0.230
Alcohols	13	0.196
Aldehydes	3	0.117
Alkanes	13	0.234
Alkenes	5	0.085
Amides	4	0.483
Amines	5	0.138
Aromatics	13	0.313
Esters	4	0.055
Ethers	4	0.241
Heterorings	5	0.122
Ketones	5	0.160
Sulfides	4	0.115
Thiols	4	0.095

Supplementary Table 7 | Release-artifact hashes

file	sha256	bytes
MODEL_CARD.md	a20bd031f5b85ba4f8fea701fa74f3d71f092ce79abb0c19b517711ff8ecfbde	1617
abraham_teacher.joblib	8c8a6365e7162dbe6ae445ce4004928ed86b2669d3e7a70d3d054f143ca97554	87055391
combisolv_qm.pt	2950be1232e2f868136e3388cf139e671d4196922ece0d3c7043d9dadece8b40	37452324
confsolv_teacher.joblib	a71487e53d5085e897dad02e62e95562933bc47262145fbf44286a04b70b0d	5729365
head.joblib	fdc88cbae31696f254d85eb427b3c53a4811f29b3290d605b4f1ea66ebb95790	29424285
implicit_teacher.joblib	484d5e0c1efa759c8e7acbfcbdd87e218b93ae239e7f35303fe4cd7e310910f8	36316890
model_card.json	0f78177a701320a5aad77c45304ea73d5ca4799fb91e4f4fd01d0bb83f8e1244	1745
mol_solv_smd.pt	35601030f030af72a9e6990bf8003df9d481858b27994b3fc70c81d605d130be	37226928
openff_teacher.joblib	55b7d31a293dd61fe57f505e015775d7d86f53dca69980d398843706d58bb9b	48211661

Supplementary Table 8 | Runtime measurements

Measurement	Value
cold single molecule	15.312 s
warm single molecule median	15.294 s
warm single molecule p95	15.322 s
batch 32 median	15.824 s
batch 32 per molecule	0.494 s

Supplementary Table 9 | Equal-weight ARROW sensitivity

All error measures are in kcal mol⁻¹.

Method	N	MAE	RMSE	Median absolute error
Matched structure-only	85	0.30977	0.59585	0.14972
Full SolvAI	85	0.20642	0.32627	0.11653

Supplementary Table 10 | External validation

All error measures are in kcal mol⁻¹; the strict 97 are a subset of the 220.

Cohort	Method	N	MAE	RMSE	Median absolute error
Endpoint-disjoint	Matched structure-only	220	1.53165	2.30974	0.96182
Endpoint-disjoint	Full SolvAI	220	1.15255	1.57723	0.84508
Strict response-source-disjoint	Matched structure-only	97	2.13830	3.08051	1.10237
Strict response-source-disjoint	Full SolvAI	97	1.53560	1.98792	1.24392

Supplementary Table 11 | Molecular-model comparisons

All error measures are in kcal mol⁻¹. For MoLFormer and Uni-Mol, “+ responses” retains the structural descriptors and adds the 15 response descriptors; it does not replace the structural input.

Set	Model	<i>N</i>	MAE	RMSE	Median AE
ARROW-85	CheMeleon (fine-tuned)	85	0.4967	0.7473	0.3251
ARROW-85	Computation-only + structure	85	0.2339	0.3936	0.1400
ARROW-85	D-MPNN (from scratch)	85	0.5590	0.8300	0.4194
ARROW-85	MoLFormer + responses	85	0.2342	0.3671	0.1252
ARROW-85	MoLFormer + structure	85	0.3481	0.5984	0.1673
ARROW-85	SolvAI	85	0.2022	0.3176	0.1169
ARROW-85	Structural descriptors	85	0.3033	0.5861	0.1592
ARROW-85	Uni-Mol + responses	85	0.2204	0.3465	0.1388
ARROW-85	Uni-Mol + structure	85	0.3344	0.5818	0.1860
External-220	CheMeleon (fine-tuned)	220	1.2716	1.8674	0.8423
External-220	Computation-only + structure	220	1.0953	1.4961	0.7768
External-220	D-MPNN (from scratch)	220	1.3013	1.6348	1.1078
External-220	MoLFormer + responses	220	1.1738	1.5743	0.9621
External-220	MoLFormer + structure	220	1.5939	2.3404	1.1811
External-220	SolvAI	220	1.1526	1.5772	0.8451
External-220	Structural descriptors	220	1.5316	2.3097	0.9618
External-220	Uni-Mol + responses	220	1.1930	1.6038	0.9377
External-220	Uni-Mol + structure	220	1.5777	2.3311	1.1301
Strict-97	CheMeleon (fine-tuned)	97	1.7102	2.3769	1.1040
Strict-97	Computation-only + structure	97	1.4216	1.8709	1.1012
Strict-97	D-MPNN (from scratch)	97	1.5584	1.9131	1.2295
Strict-97	MoLFormer + responses	97	1.5635	1.9776	1.3206
Strict-97	MoLFormer + structure	97	2.2247	3.1159	1.3580
Strict-97	SolvAI	97	1.5356	1.9879	1.2439
Strict-97	Structural descriptors	97	2.1383	3.0805	1.1024
Strict-97	Uni-Mol + responses	97	1.5889	2.0088	1.3365
Strict-97	Uni-Mol + structure	97	2.2053	3.0961	1.2184

Supplementary Table 12 | Paired molecular-model comparisons

Differences are candidate minus reference MAE in kcal mol⁻¹; negative values favor the candidate. Intervals use 100,000 molecule resamples and are descriptive, without multiplicity correction. The strict set is a subset of the external set. Model labels follow Supplementary Table 11; “+ responses” also retains structure.

Set	Contrast	Δ MAE	95% lower	95% upper
ARROW-85	SolvAI – Structural descriptors	−0.1011	−0.2151	−0.0203
ARROW-85	SolvAI – Computation-only + structure	−0.0316	−0.0625	−0.0064
ARROW-85	SolvAI – D-MPNN (from scratch)	−0.3568	−0.4742	−0.2498
ARROW-85	SolvAI – CheMeleon (fine-tuned)	−0.2944	−0.3887	−0.2109
ARROW-85	SolvAI – MoLFormer + structure	−0.1459	−0.2493	−0.0600
ARROW-85	SolvAI – MoLFormer + responses	−0.0319	−0.0541	−0.0110
ARROW-85	SolvAI – Uni-Mol + structure	−0.1322	−0.2362	−0.0492
ARROW-85	SolvAI – Uni-Mol + responses	−0.0181	−0.0346	−0.0018
ARROW-85	MoLFormer + responses – MoLFormer + structure	−0.1139	−0.2095	−0.0322
ARROW-85	Uni-Mol + responses – Uni-Mol + structure	−0.1141	−0.2195	−0.0288
ARROW-85	Computation-only + structure – Structural descriptors	−0.0695	−0.1912	0.0223
External-220	SolvAI – Structural descriptors	−0.3791	−0.5169	−0.2482
External-220	SolvAI – Computation-only + structure	0.0573	0.0230	0.0927
External-220	SolvAI – D-MPNN (from scratch)	−0.1487	−0.2733	−0.0208
External-220	SolvAI – CheMeleon (fine-tuned)	−0.1190	−0.2165	−0.0228
External-220	SolvAI – MoLFormer + structure	−0.4414	−0.5814	−0.3077
External-220	SolvAI – MoLFormer + responses	−0.0212	−0.0433	0.0008
External-220	SolvAI – Uni-Mol + structure	−0.4251	−0.5649	−0.2928
External-220	SolvAI – Uni-Mol + responses	−0.0405	−0.0597	−0.0210
External-220	MoLFormer + responses – MoLFormer + structure	−0.4201	−0.5619	−0.2862
External-220	Uni-Mol + responses – Uni-Mol + structure	−0.3847	−0.5247	−0.2522
External-220	Computation-only + structure – Structural descriptors	−0.4364	−0.5823	−0.2973
Strict-97	SolvAI – Structural descriptors	−0.6027	−0.8628	−0.3586
Strict-97	SolvAI – Computation-only + structure	0.1140	0.0584	0.1724
Strict-97	SolvAI – D-MPNN (from scratch)	−0.0228	−0.2694	0.2313
Strict-97	SolvAI – CheMeleon (fine-tuned)	−0.1746	−0.3437	−0.0117
Strict-97	SolvAI – MoLFormer + structure	−0.6891	−0.9478	−0.4439
Strict-97	SolvAI – MoLFormer + responses	−0.0279	−0.0640	0.0083
Strict-97	SolvAI – Uni-Mol + structure	−0.6697	−0.9251	−0.4290
Strict-97	SolvAI – Uni-Mol + responses	−0.0533	−0.0822	−0.0247
Strict-97	MoLFormer + responses – MoLFormer + structure	−0.6612	−0.9282	−0.4102
Strict-97	Uni-Mol + responses – Uni-Mol + structure	−0.6164	−0.8764	−0.3722
Strict-97	Computation-only + structure – Structural descriptors	−0.7167	−0.9938	−0.4540

Supplementary Table 13 | Extended endpoint comparisons

All errors are in kcal mol⁻¹. Full predictions average five seeds for neural and TabPFN fits and three for the tree controls. The last column recomputes MAE using only the three shared seeds. No-response controls inherit selected recipes. Count- and width-adaptive rows are selection policies, not additional architectures. The three reported cohorts are not independent: Strict-97 is nested within External-220.

Set	Endpoint	Input	MAE	RMSE	Median	3-seed MAE
ARROW-85	ExtraTrees	No response	0.3033	0.5861	0.1592	0.3033
ARROW-85	ExtraTrees	Response	0.2022	0.3176	0.1169	0.2022
ARROW-85	Residual MLP	No response	0.3581	0.6285	0.2086	0.3592
ARROW-85	Residual MLP	Response	0.3403	0.5384	0.2039	0.3308
ARROW-85	Dual branch	No response	0.3444	0.6234	0.1522	0.3478
ARROW-85	Dual branch	Response	0.2884	0.4574	0.1836	0.2893
ARROW-85	Ridge + neural residual	No response	0.3864	0.7521	0.2181	0.3822
ARROW-85	Ridge + neural residual	Response	0.3360	0.5268	0.2147	0.3343
ARROW-85	Message passing	No response	0.3222	0.6459	0.1412	0.3243
ARROW-85	Message passing	Response	0.2877	0.5178	0.1540	0.2956
ARROW-85	MoLFormer + neural head	No response	0.4647	0.7020	0.2792	0.4648
ARROW-85	MoLFormer + neural head	Response	0.2729	0.4241	0.1741	0.2883
ARROW-85	TabM	No response	0.3367	0.6415	0.1619	0.3415
ARROW-85	TabM	Response	0.2910	0.4608	0.1791	0.2901
ARROW-85	TabPFN-3.5	No response	0.2179	0.4384	0.1062	0.2166
ARROW-85	TabPFN-3.5	Response	0.1992	0.3916	0.0786	0.1965
ARROW-85	MoLFormer + ExtraTrees	No response	0.3481	0.5984	0.1673	0.3481
ARROW-85	MoLFormer + ExtraTrees	Response	0.2342	0.3671	0.1252	0.2342
ARROW-85	Count-adaptive selection	No response	0.3222	0.6459	0.1412	0.3243
ARROW-85	Count-adaptive selection	Response	0.2877	0.5178	0.1540	0.2956
ARROW-85	Width-adaptive selection	No response	0.3732	0.7292	0.1559	0.3847
ARROW-85	Width-adaptive selection	Response	0.2879	0.5011	0.1455	0.2884
External-220	ExtraTrees	No response	1.5316	2.3097	0.9618	1.5316
External-220	ExtraTrees	Response	1.1526	1.5772	0.8451	1.1526
External-220	Residual MLP	No response	1.5246	2.1985	1.0809	1.5381
External-220	Residual MLP	Response	1.4281	2.0754	0.9788	1.4499
External-220	Dual branch	No response	1.5362	2.2491	1.0459	1.5402
External-220	Dual branch	Response	1.2339	1.7405	0.9392	1.2320
External-220	Ridge + neural residual	No response	1.1313	1.6178	0.7443	1.1272
External-220	Ridge + neural residual	Response	1.1723	1.6670	0.8471	1.1684
External-220	Message passing	No response	1.3110	1.8704	0.9396	1.2773
External-220	Message passing	Response	1.1678	1.6740	0.8554	1.1443
External-220	MoLFormer + neural head	No response	1.3325	1.8425	1.0735	1.3486
External-220	MoLFormer + neural head	Response	1.1113	1.5561	0.8664	1.1332
External-220	TabM	No response	1.2203	1.6453	0.9585	1.2177
External-220	TabM	Response	1.1317	1.5958	0.8627	1.1542
External-220	TabPFN-3.5	No response	1.1311	1.5627	0.8500	1.1534
External-220	TabPFN-3.5	Response	1.1339	1.6386	0.8355	1.1343
External-220	MoLFormer + ExtraTrees	No response	1.5939	2.3404	1.1811	1.5939
External-220	MoLFormer + ExtraTrees	Response	1.1738	1.5743	0.9621	1.1738
External-220	Count-adaptive selection	No response	1.1348	1.6092	0.7699	1.1349
External-220	Count-adaptive selection	Response	1.1479	1.6568	0.8179	1.1349
External-220	Width-adaptive selection	No response	1.3207	1.8695	0.9498	1.3250
External-220	Width-adaptive selection	Response	1.0904	1.5281	0.8664	1.0751

Set	Endpoint	Input	MAE	RMSE	Median	3-seed MAE
Strict-97	ExtraTrees	No response	2.1383	3.0805	1.1024	2.1383
Strict-97	ExtraTrees	Response	1.5356	1.9879	1.2439	1.5356
Strict-97	Residual MLP	No response	2.1525	2.9344	1.5080	2.1652
Strict-97	Residual MLP	Response	2.0162	2.7645	1.4467	2.0605
Strict-97	Dual branch	No response	2.1889	3.0245	1.4705	2.1903
Strict-97	Dual branch	Response	1.6165	2.1756	1.3722	1.6141
Strict-97	Ridge + neural residual	No response	1.2941	1.7894	0.8337	1.2815
Strict-97	Ridge + neural residual	Response	1.4288	1.9521	0.9264	1.4289
Strict-97	Message passing	No response	1.7152	2.3425	1.2040	1.6307
Strict-97	Message passing	Response	1.5054	2.0904	1.1651	1.4489
Strict-97	MoLFormer + neural head	No response	1.6850	2.2703	1.2297	1.7110
Strict-97	MoLFormer + neural head	Response	1.4501	1.9473	1.0978	1.4830
Strict-97	TabM	No response	1.5655	2.0315	1.2785	1.5586
Strict-97	TabM	Response	1.4540	1.9517	1.1925	1.5056
Strict-97	TabPFN-3.5	No response	1.3467	1.8596	0.9823	1.3926
Strict-97	TabPFN-3.5	Response	1.4932	2.0476	1.0358	1.4854
Strict-97	MoLFormer + ExtraTrees	No response	2.2247	3.1159	1.3580	2.2247
Strict-97	MoLFormer + ExtraTrees	Response	1.5635	1.9776	1.3206	1.5635
Strict-97	Count-adaptive selection	No response	1.2974	1.7776	0.8305	1.2966
Strict-97	Count-adaptive selection	Response	1.3812	1.9387	0.8740	1.3502
Strict-97	Width-adaptive selection	No response	1.6271	2.2899	1.1027	1.6406
Strict-97	Width-adaptive selection	Response	1.3723	1.8633	1.0467	1.3222

Supplementary Table 14 | Matched response contrasts across endpoints

Within each endpoint, response-minus-no-response MAE and descriptive 95% paired molecule-bootstrap intervals, in kcal mol⁻¹; 100,000 resamples without multiplicity adjustment. Negative values favor adding responses. The candidate always includes responses; "No response" is its matched ablation.

Set	Endpoint	Δ MAE	Lower	Upper
ARROW-85	ExtraTrees	-0.1011	-0.2151	-0.0203
ARROW-85	Residual MLP	-0.0179	-0.0543	0.0127
ARROW-85	Dual branch	-0.0559	-0.1244	0.0066
ARROW-85	Ridge + neural residual	-0.0504	-0.1403	0.0263
ARROW-85	Message passing	-0.0345	-0.0997	0.0189
ARROW-85	MoLFormer + neural head	-0.1918	-0.2925	-0.1005
ARROW-85	TabM	-0.0457	-0.1138	0.0129
ARROW-85	TabPFN-3.5	-0.0187	-0.0463	0.0072
ARROW-85	MoLFormer + ExtraTrees	-0.1139	-0.2095	-0.0322
ARROW-85	Count-adaptive selection	-0.0345	-0.0997	0.0189
ARROW-85	Width-adaptive selection	-0.0853	-0.1652	-0.0177
External-220	ExtraTrees	-0.3791	-0.5169	-0.2482
External-220	Residual MLP	-0.0965	-0.1372	-0.0575
External-220	Dual branch	-0.3023	-0.4151	-0.1960
External-220	Ridge + neural residual	0.0410	-0.0335	0.1180
External-220	Message passing	-0.1432	-0.2174	-0.0765
External-220	MoLFormer + neural head	-0.2212	-0.3096	-0.1343
External-220	TabM	-0.0886	-0.1324	-0.0441
External-220	TabPFN-3.5	0.0028	-0.0992	0.1122
External-220	MoLFormer + ExtraTrees	-0.4201	-0.5619	-0.2862
External-220	Count-adaptive selection	0.0132	-0.0642	0.0934
External-220	Width-adaptive selection	-0.2303	-0.3186	-0.1451
Strict-97	ExtraTrees	-0.6027	-0.8628	-0.3586
Strict-97	Residual MLP	-0.1362	-0.2132	-0.0625
Strict-97	Dual branch	-0.5724	-0.7824	-0.3750
Strict-97	Ridge + neural residual	0.1346	0.0068	0.2699
Strict-97	Message passing	-0.2098	-0.3468	-0.0889
Strict-97	MoLFormer + neural head	-0.2349	-0.3810	-0.0869
Strict-97	TabM	-0.1116	-0.1827	-0.0409
Strict-97	TabPFN-3.5	0.1465	-0.0470	0.3611
Strict-97	MoLFormer + ExtraTrees	-0.6612	-0.9282	-0.4102
Strict-97	Count-adaptive selection	0.0838	-0.0531	0.2277
Strict-97	Width-adaptive selection	-0.2548	-0.3934	-0.1184

Supplementary Table 15 | Exploratory size holdout

The same 205 held-out molecules and 1,160 training labels are used for every row. All fits average three seeds. Errors are in kcal mol⁻¹; the last column is each endpoint’s response-minus-no-response MAE, not a cross-endpoint ranking. Cross-endpoint paired intervals are in Supplementary Data 6, `size_diagnostic_comparisons.csv`. These development labels were available in earlier random-fold evaluation.

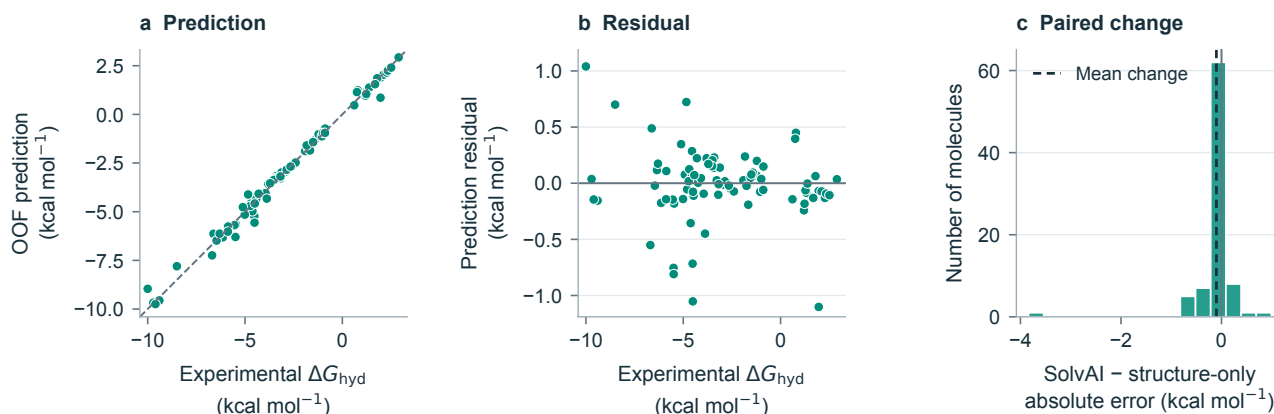
Endpoint	Response MAE	No-response MAE	Δ MAE
ExtraTrees	1.7716	2.2459	-0.4743
Residual MLP	1.8130	1.9921	-0.1791
Dual branch	1.5600	1.9753	-0.4154
Ridge + neural residual	1.3913	1.4622	-0.0709
Message passing	1.7293	1.9303	-0.2010
MoLFormer + neural head	1.6091	2.0165	-0.4074
TabM	1.3886	1.6891	-0.3006
TabPFN-3.5	1.3185	1.9586	-0.6402
Count-adaptive selection	1.3369	1.4470	-0.1101
Width-adaptive selection	1.5215	1.9901	-0.4686

Supplementary Table 16 | Native TabPFN interval coverage

Intervals are central quantiles of the equal five-model predictive mixture. Widths are in kcal mol⁻¹; no external calibration was applied.

Set	Nominal	Covered	Observed	Mean width	Median width
External-220	80%	128/220	58.2%	2.605	1.634
External-220	95%	176/220	80.0%	4.605	2.860
Strict-97	80%	54/97	55.7%	3.539	2.728
Strict-97	95%	74/97	76.3%	6.200	4.678

Supplementary Figures

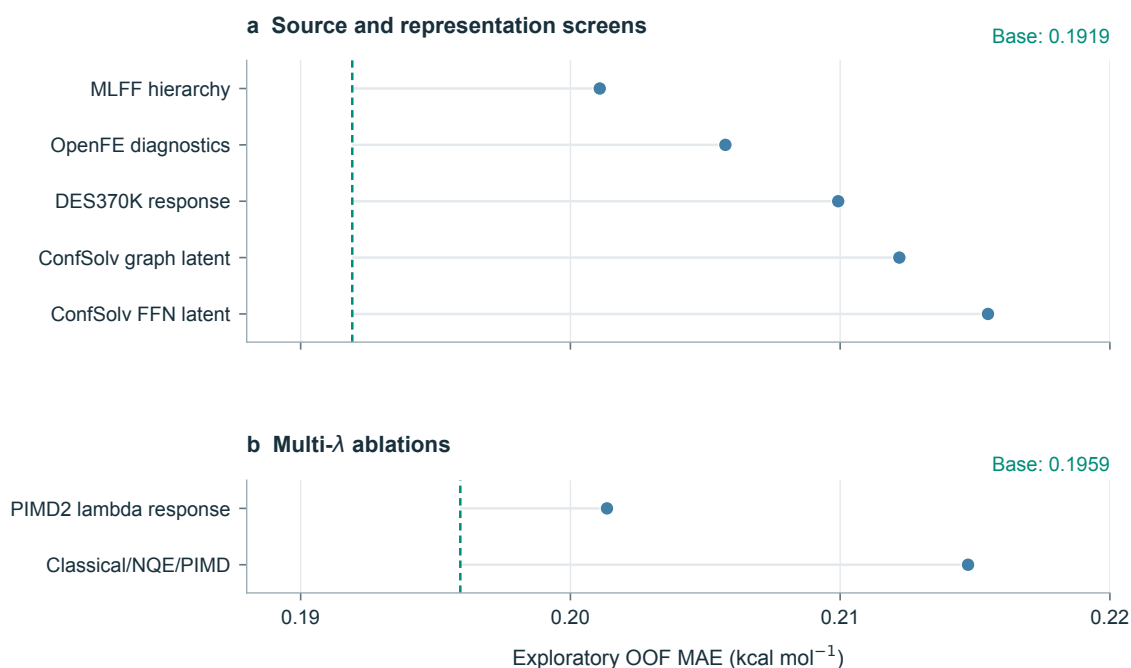


Supplementary Figure 1 | Molecule-level prediction and residuals. **a**, Confirmatory response-augmented ExtraTrees OOF prediction against experiment. **b**, Signed residual (prediction minus experiment) against experimental hydration free energy. **c**, Distribution of the paired absolute-error change, responses minus the matched structure-only endpoint; negative values favor responses.

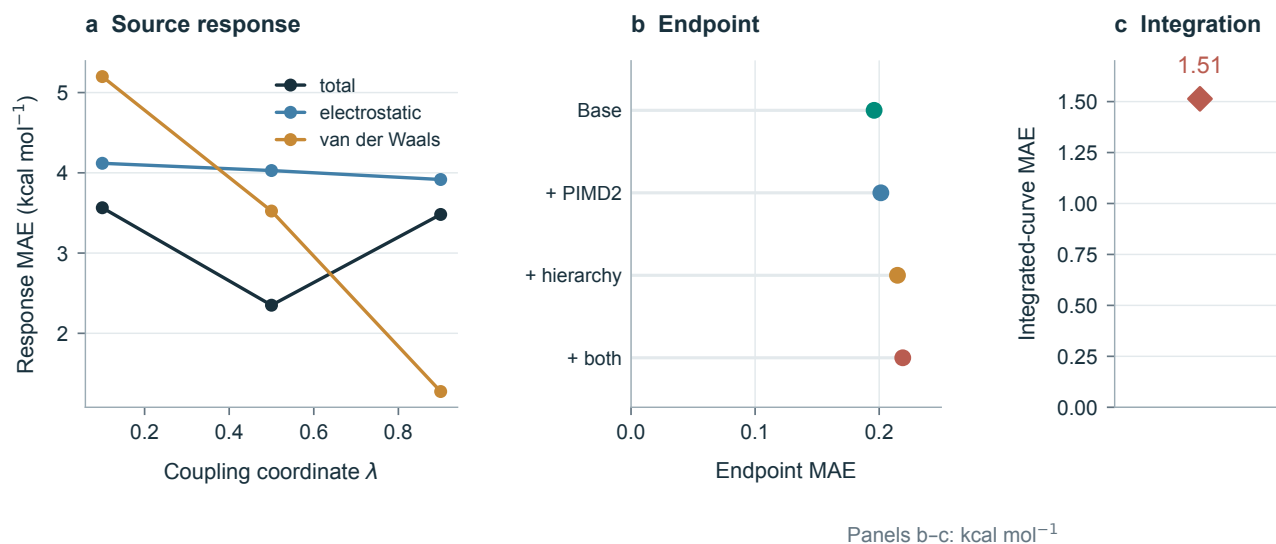
Source	Counting unit	Before	Removed	Retained
CombiSolv-QM	structures	3,961	2	3,959
MolSolv	unique structures	350,391	32	350,359
ConfSolv	usable connectivities	17,851	22	17,829
Endpoint labels	connectivities	1,280	0	1,280

Counts retain their source-specific units; they are not pooled.

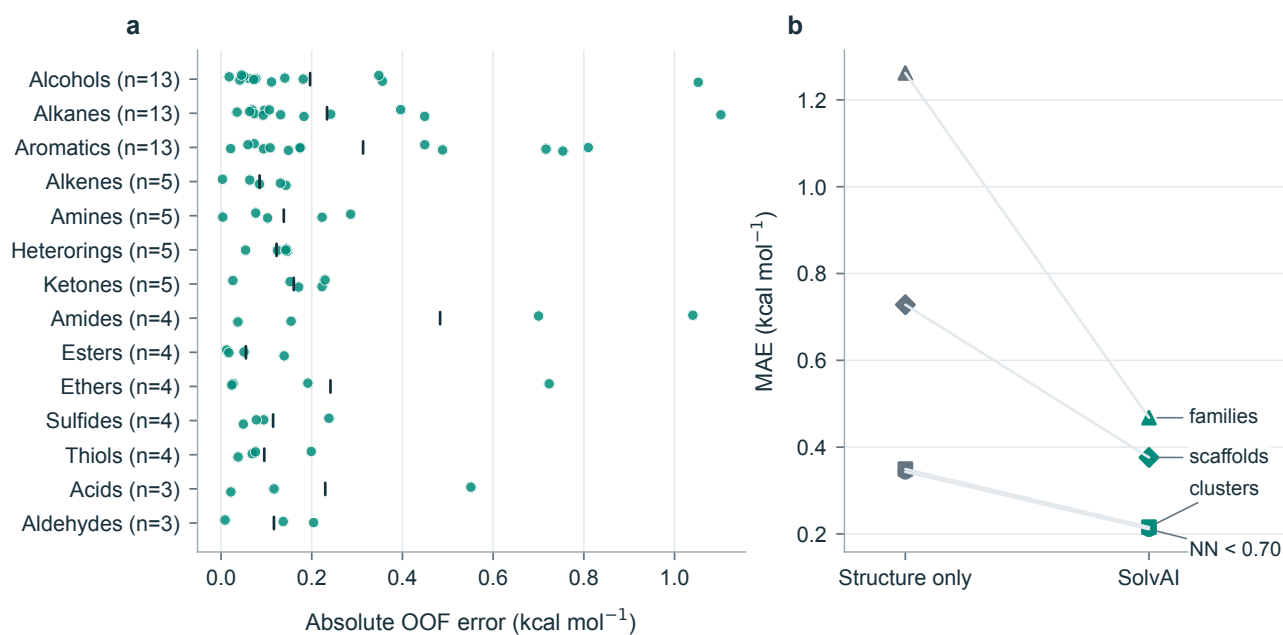
Supplementary Figure 2 | Source-specific provenance and standardized identity exclusion. Counts before exclusion, removed counts and retained counts are shown in their native source units. The audit removed standardized benchmark equivalents before the confirmatory surrogate refits; no endpoint experimental label was removed.



Supplementary Figure 3 | Alternative physical-supervision routes. Exploratory fixed screens of additional diagnostic, latent and hierarchy features. **a**, Source and representation screens use the SMD(water) plus ConfSolv response base (0.1919 kcal mol⁻¹). **b**, Multi- λ ablations use their own fixed baseline (0.1959 kcal mol⁻¹). Each dashed line marks the corresponding control; gray segments connect that control to each screened alternative. These experiments were not used to select the confirmatory controls. OpenFE denotes the Open Free Energy toolkit; MLFF denotes machine-learning force field; DES370K denotes the roughly 370,000-geometry intermolecular-interaction dataset; FFN denotes feed-forward network.



Supplementary Figure 4 | Multi- λ response experiment. **a**, OOF response-head error at three coupling states. **b**, Matched endpoint consequences. **c**, Separately scaled error obtained by direct integration of the predicted curve with fold-local affine calibration (diamond).



Supplementary Figure 5 | Molecular and chemical-separation errors. **a**, Individual fixed-partition response-augmented ExtraTrees errors grouped by diagnostic chemical family; ticks show family means and labels give sample sizes. **b**, Matched structure-only and response-augmented ExtraTrees errors under four global separation regimes.

a Keep test labels outside fitting and selection

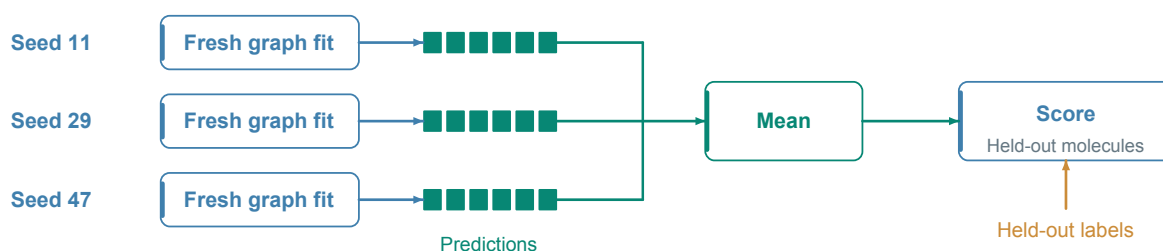
Outer-training labels



The 15% inner split is selected separately within each label source.

b Average independent seeds before scoring

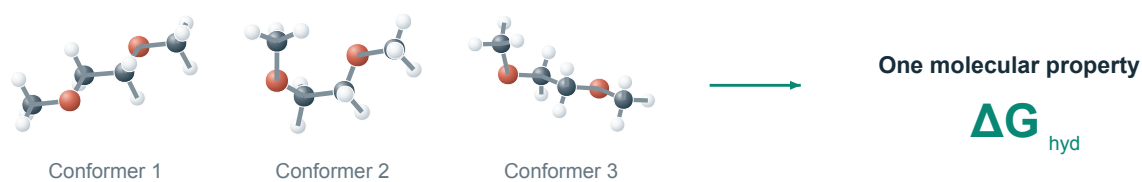
Repeat selection and refit for every seed



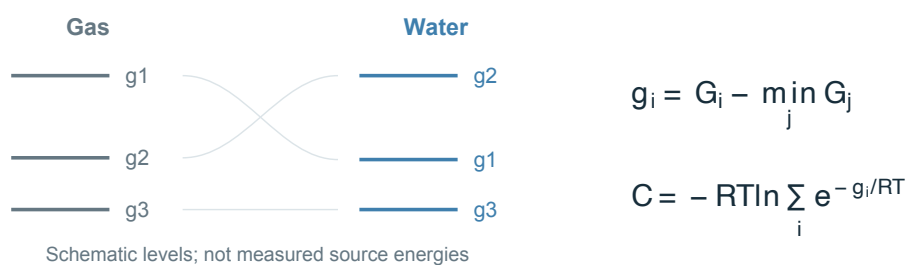
Five ARROW folds + one external fit; three seeds × two direct graph models = 36 runs.

Supplementary Figure 6 | Select graph-model duration without using held-out labels. **a**, Within each outer-training set and seed, a deterministic 15% of each label source is reserved for inner validation. The epoch with the lowest unweighted validation MAE is selected, subject to an 80-epoch cap and patience 12. A model reinitialized from its original random or CheMeleon starting weights is then refitted on all outer-training labels for that many epochs, with a newly fitted training scaler. **b**, Selection and refitting are repeated for seeds 11, 29 and 47. Their predictions are averaged before comparison with held-out labels. The five ARROW folds and one all-training external fit give 36 seed-level runs across the scratch D-MPNN and fine-tuned CheMeleon models. The colored vectors represent predictions for held-out molecules, not new training inputs.

a Many conformers, one equilibrium hydration target



b Summarize the supplied ensembles



c Predict six summaries from molecular structure



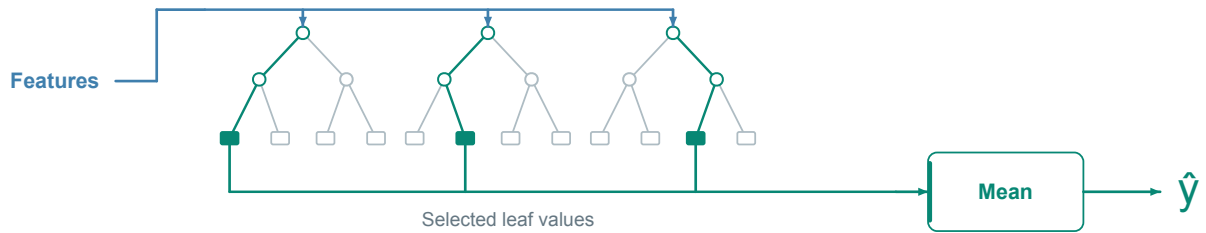
s_i : conformer solvation energy; i_0 : lowest-gas-energy conformer.

$\Delta g_i = g_i^{\text{water}} - g_i^{\text{gas}}$. Each phase uses its own energy minimum.

Supplementary Figure 7 | Conformer-resolved sources teach a molecular equilibrium response. **a**, Three illustrative conformers of 1,2-dimethoxyethane share one molecular equilibrium hydration target; they do not have separate experimental endpoint labels. **b**, Gas and water conformer energies are normalized separately to their phase-specific minima, giving g_i . The correction $C = -RT \ln \sum_i \exp(-g_i/RT)$ summarizes the supplied ensemble. Energy levels and their connecting lines are schematic, not measured ConfSolv energies. **c**, The six predicted summaries are the gas and water corrections, ensemble hydration minus the solvation energy of the lowest-gas-energy conformer, the population standard deviation of conformer solvation energies, and the arithmetic mean and population standard deviation of Δg_i . All energies are in kcal mol⁻¹. These summaries do not assert complete conformational sampling, and no conformer ensemble is regenerated at deployment.

a ExtraTrees

Partition feature space; average one stored leaf value from every tree.

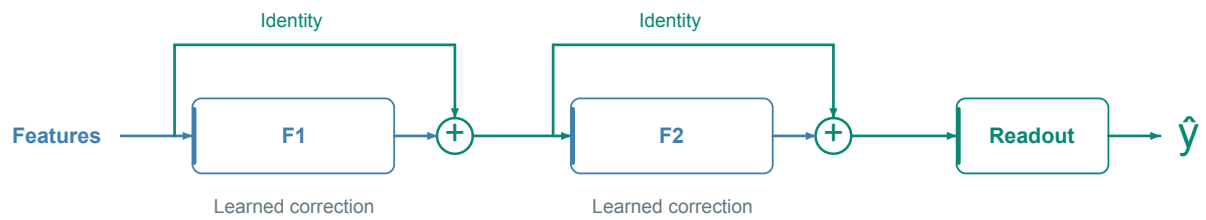


ARROW MAE 0.202 · Matched and shuffled-response controls

A finite range of leaf values bounds the prediction.

b Residual MLP

Learn nonlinear corrections along an identity-preserving pathway.

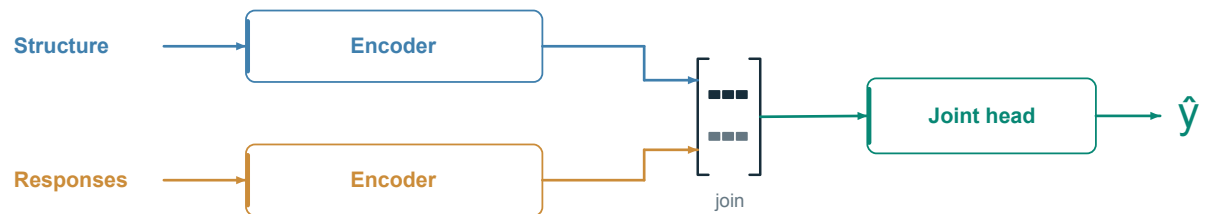


ARROW MAE 0.340 · Tests a direct neural replacement

A flexible function alone does not recover the tree accuracy.

c Dual structure / response branches

Learn a representation for each input group before combining them.



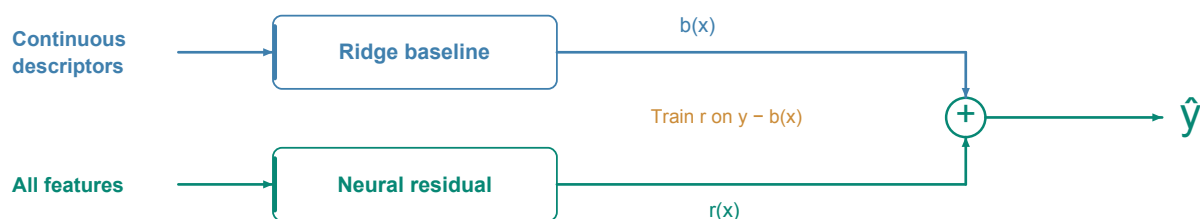
Size-held-out MAE 1.560 · Refitted ExtraTrees: 1.772

Separate branches do not impose physical additivity.

Supplementary Figure 8 | Local averaging and neural feature combination. **a**, ExtraTrees selects a leaf in each randomized tree and averages its stored value. **b**, A residual MLP learns corrections alongside identity paths. **c**, Separate structural and response branches are concatenated before a joint head. Illustrations explain mechanisms, not exact layer widths or tree counts. Numerical callouts refer to the evaluations in Table 2, not theoretical properties of the architectures. All mechanism diagrams use editable vector geometry; they are explanatory schematics, not data-derived plots.

a Ridge + neural residual

Combine a continuous baseline with a learned correction.



Size-held-out MAE 1.391 · No tree-leaf ceiling in the correction

A linear trend is not a validated asymptotic free-energy law.

b Size-sensitive message passing

Count and aggregate atom states, not only fragment presence.

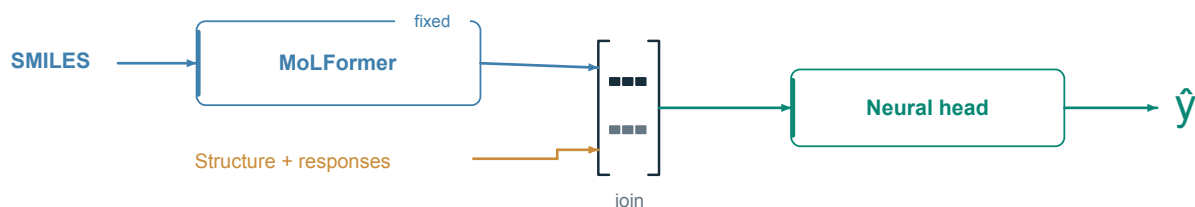


External MAE 1.168 · Without responses: 1.311

Shown: sum / mean / count readout; an atomic-plus-global variant was also tested.

c Frozen MoLFormer + neural head

Reuse a pretrained sequence representation alongside response information.



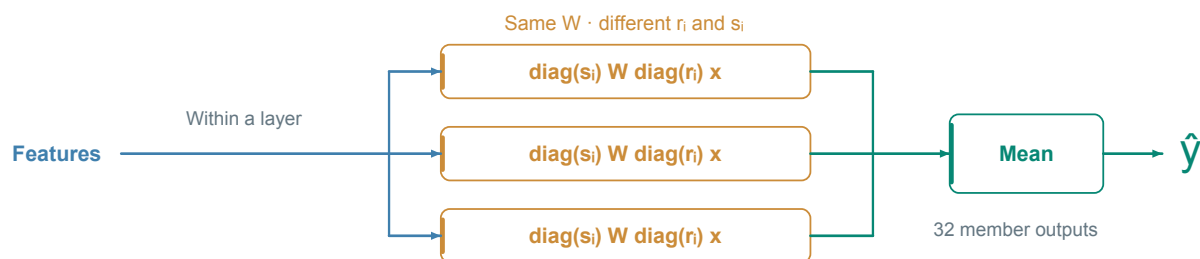
External MAE 1.111 · Responses help in all three cohorts

The encoder stays fixed; the endpoint learns from hydration labels.

Supplementary Figure 9 | Baselines, graph aggregation and pretrained encodings. **a**, Ridge predicts a continuous baseline and a network learns the remaining target, $y - b(x)$, during training; inference adds the baseline and predicted residual without access to the query's target. **b**, One graph variant combines summed and mean atom states with atom-count information; a separate atomic-plus-global variant is also tested. The summaries are concatenated, not numerically equated or treated as independent free-energy contributions. **c**, MoLFormer is frozen, while the downstream head learns to combine its representation with structural and response information. Schematics omit layer detail; all source surrogates remain fixed.

a TabM: a parameter-sharing ensemble

Use member-specific factors within layers that share their main weights.

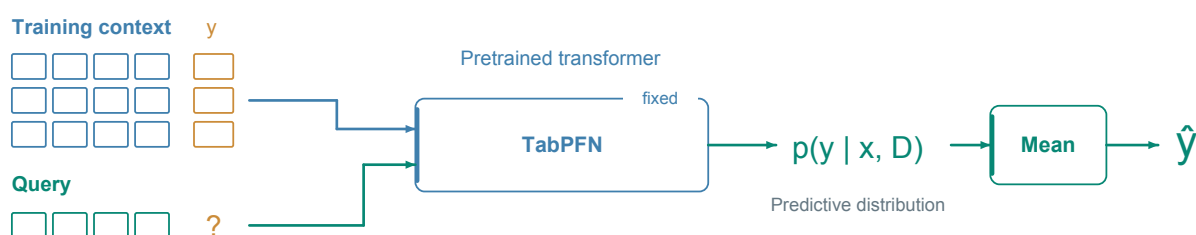


External MAE 1.132 · Size-held-out MAE 1.389

32 members share parameters; these are not 32 independent training runs.

b TabPFN-3.5: prediction from context

Condition a pretrained transformer on eligible labelled training rows.

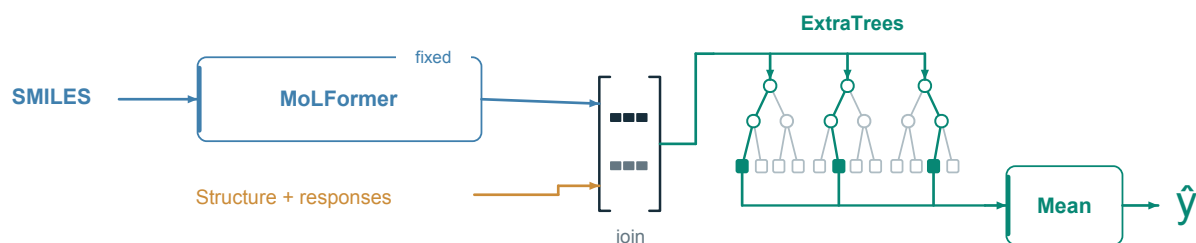


ARROW MAE 0.199 · Size-held-out MAE 1.318

Native intervals under-cover; input recipes differ across partitions.

c Frozen MoLFormer + ExtraTrees

Change the representation while keeping the tree endpoint family.

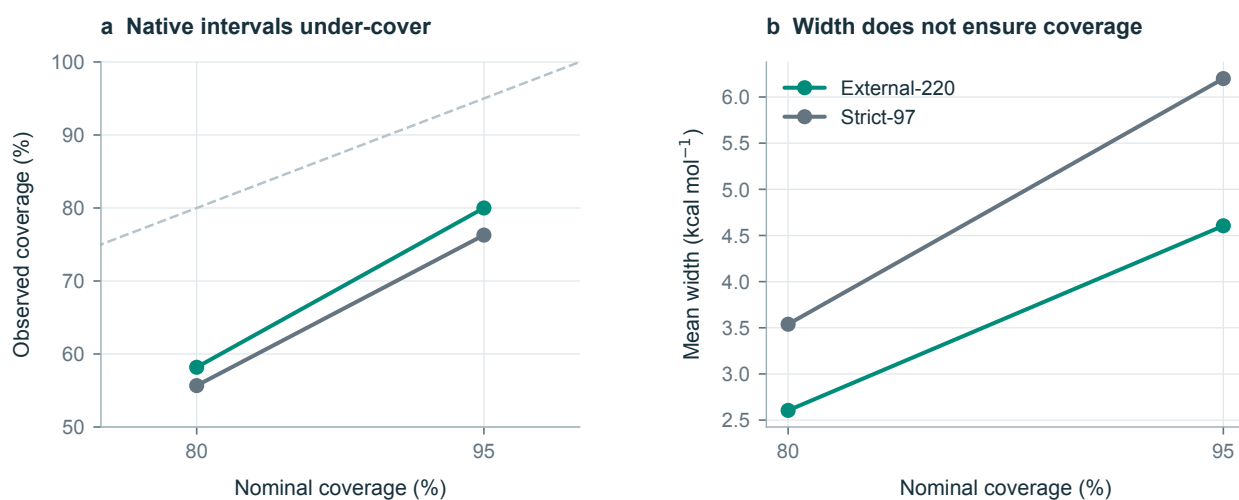


ARROW MAE 0.234 · Without responses: 0.348

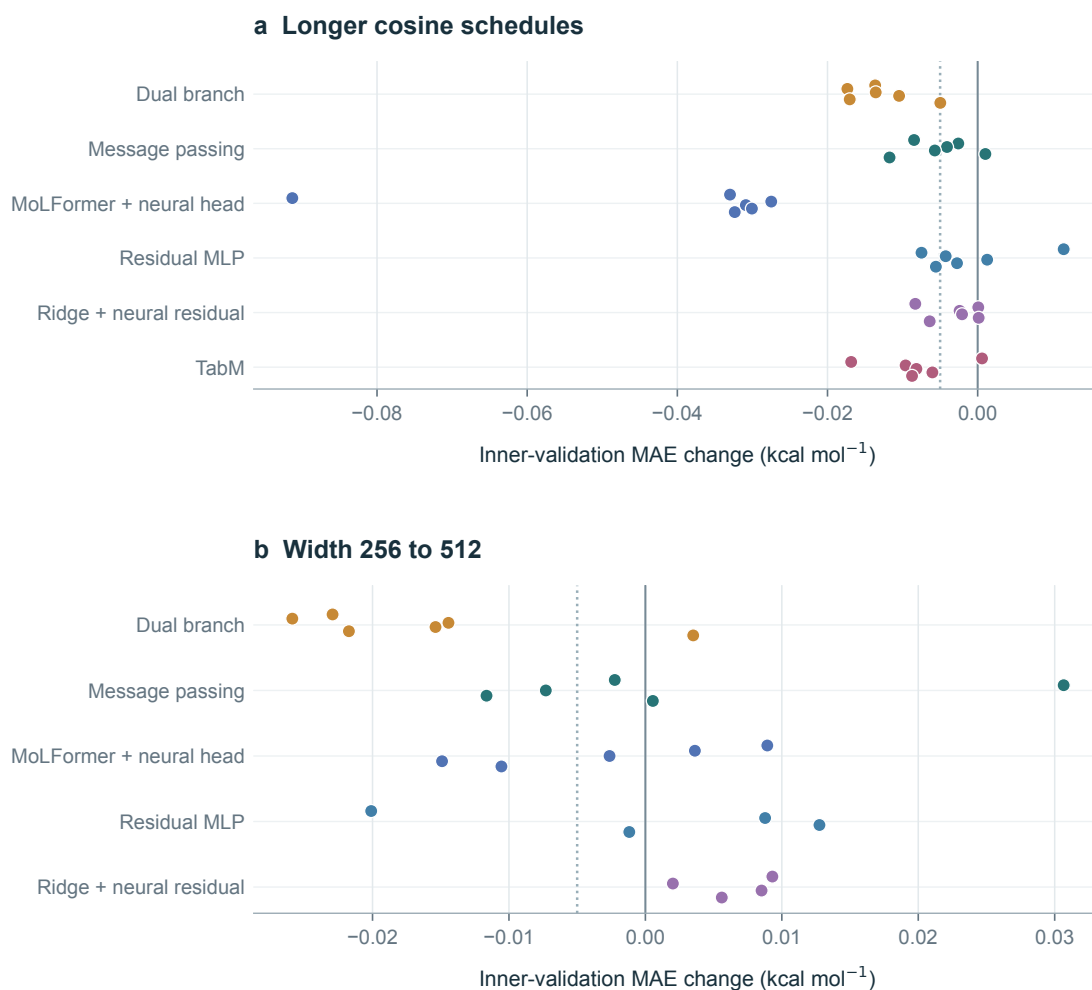
The endpoint remains bounded by its stored leaf values.

Supplementary Figure 10 | Shared ensembles, in-context inference and a representation control.

a, TabM shares weight matrices while member-specific input/output modulations within its layers produce 32 predictions. Three paths are shown for clarity; the mean is used. **b**, TabPFN conditions on labelled training rows and an unlabelled query using pretrained weights. No query label is supplied and no gradient fine-tuning is done here. Feature recipes vary by partition (S19). **c**, Frozen MoLFormer features are combined with structural and response features before an ExtraTrees endpoint, isolating a representation change while retaining leaf-value averaging. Three representative trees are shown, not the full ensemble.



Supplementary Figure 11 | Native predictive intervals require calibration. **a**, Empirical versus nominal coverage for the equal five-model TabPFN mixture; the dashed diagonal marks nominal coverage. **b**, Corresponding mean interval widths. Strict-97 is nested in External-220. These intervals are evaluated without external calibration, and their undercoverage is not represented by seed dispersion. Connecting lines guide the eye between the two assessed levels.



Supplementary Figure 12 | Training duration and capacity challenges. **a**, Mean three-split inner-validation changes at seed 47 for the 1,200-epoch cosine challenge relative to the ordinary schedule, one dot per outer partition and family. **b**, Corresponding changes when width is expanded from 256 to 512 for qualifying configurations. Negative favors the challenge; the dotted line marks -0.005 kcal mol⁻¹. Retention also requires improvement in at least two of the three pairs. These are inner-validation scores, not test results, and the duration challenge also changes the learning-rate schedule.